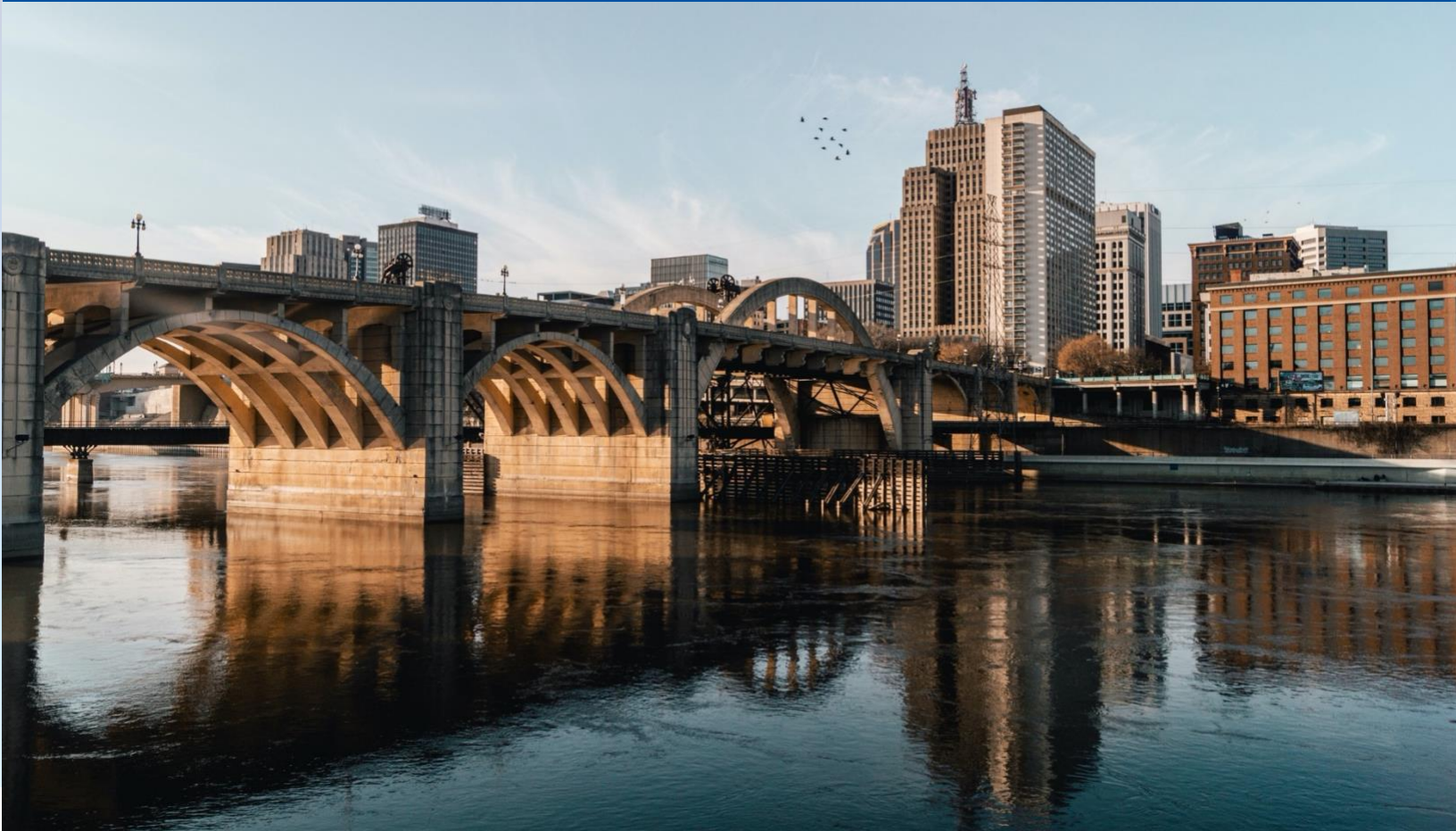




VMT Reduction Mode Shift Study



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of NORTH CAROLINA
at CHAPEL HILL



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Executive Summary

In this study, we consider the potential to reduce vehicle miles traveled (VMT) in the Twin Cities region that might be achieved by incentivizing people to switch from traveling in cars to walking, biking or riding transit. We do so by **estimating the maximum feasible shift of car trips to alternative modes given existing land-use and travel patterns**, as observed in 380,000 real-world trips reported in the 2019 and 2021 Travel Behavior Inventory (TBI) surveys. For each observed car trip, we calculate the best available walk path, bike path, and transit path, considering the quality of the available infrastructure and transit service. We consider an option to be *feasible* if we observe a substantial number (5% or more) of people using that mode under similar circumstances, and if the traveler has sufficient time to complete the trip within their travel schedule. Further, we consider a feasible option to be *competitive* if its travel time is within 15 minutes of the car trip it would replace.

Our analysis also examines which modes can absorb the largest mode shift, as summarized in Figure 1. Our results show that walking is a competitive alternative for 13% of car trips. However, because walking is most competitive for short trips, converting these trips to walk would only reduce VMT by 1%. Biking is a competitive alternative for 17% of car trips, comprising 3% of VMT, and transit is competitive for 6% of car trips, comprising 1% of VMT. Biking and transit are both feasible options for much larger share of VMT than walking, but there is a larger gap between feasible and competitive. This gap highlights the potential to improve the competitiveness of these modes by enabling more direct bike routes or faster and more frequent transit service.

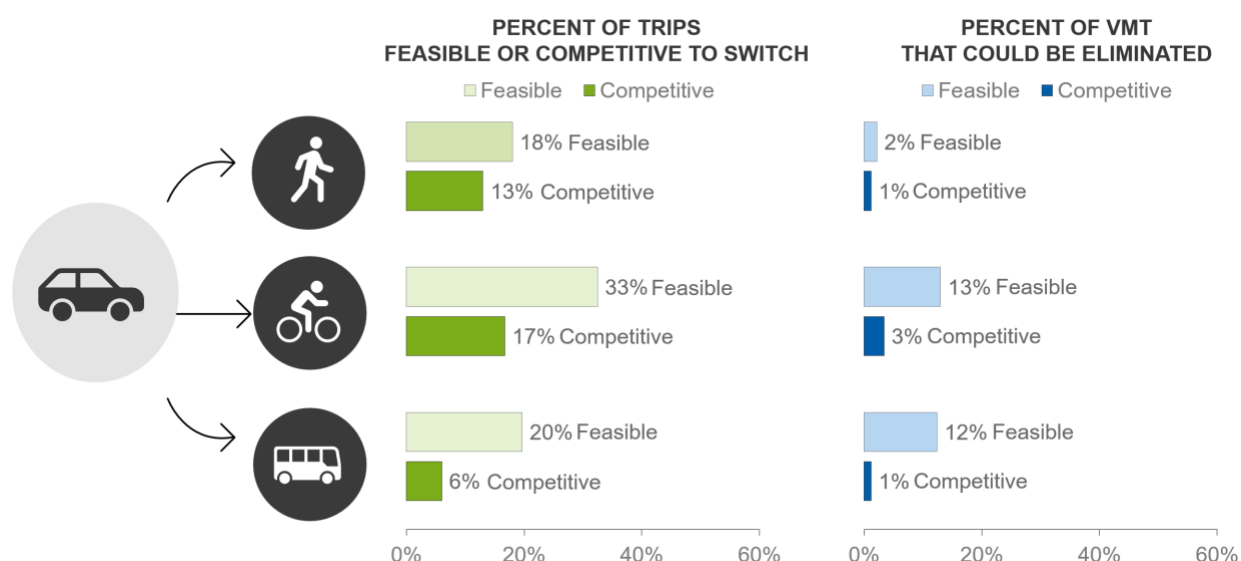


Figure 1 Potential to reduce vehicle trips and VMT by shifting car trips to walking, biking or transit

For some trips, more than one alternative mode is feasible, and in those cases we consider the competitiveness of the fastest available mode. **Overall, in the Twin Cities region today, 25% of car trips could shift to walk, bike or transit with travel times within 15 minutes of driving, and these trips represent 4.5% of VMT.**

This analysis shows that given existing travel patterns, even if everyone who could competitively switch to walking, biking and transit did switch, it would only achieve a quarter of the targeted 20% per capita VMT reduction. Therefore, policy-makers may wish to consider transportation infrastructure and service changes to further encourage mode shifts. To provide a starting point for those discussions, we evaluated the potential of six very broad scenarios to increase the competitiveness of alternative modes:

- Walk infrastructure: it is safe and comfortable to walk everywhere it is legal to do so;
- Bike infrastructure: it is safe and comfortable to bike everywhere it is legal to do so;
- E-bikes: additionally, everyone has an e-bike;
- Transit frequency: transit frequency is doubled for all routes;
- Highway speed: speed limits are reduced based on NACTO guidelines; and
- All of the above.

Figure 2 shows the results of this scenario analysis, considering the percent of car trips and VMT for which the fastest feasible alternative in each scenario is within 15 minutes of the car travel time. The results show that among the individual scenarios, the e-bike scenario provides a competitive alternative for the greatest share of VMT, followed by the bike scenario.

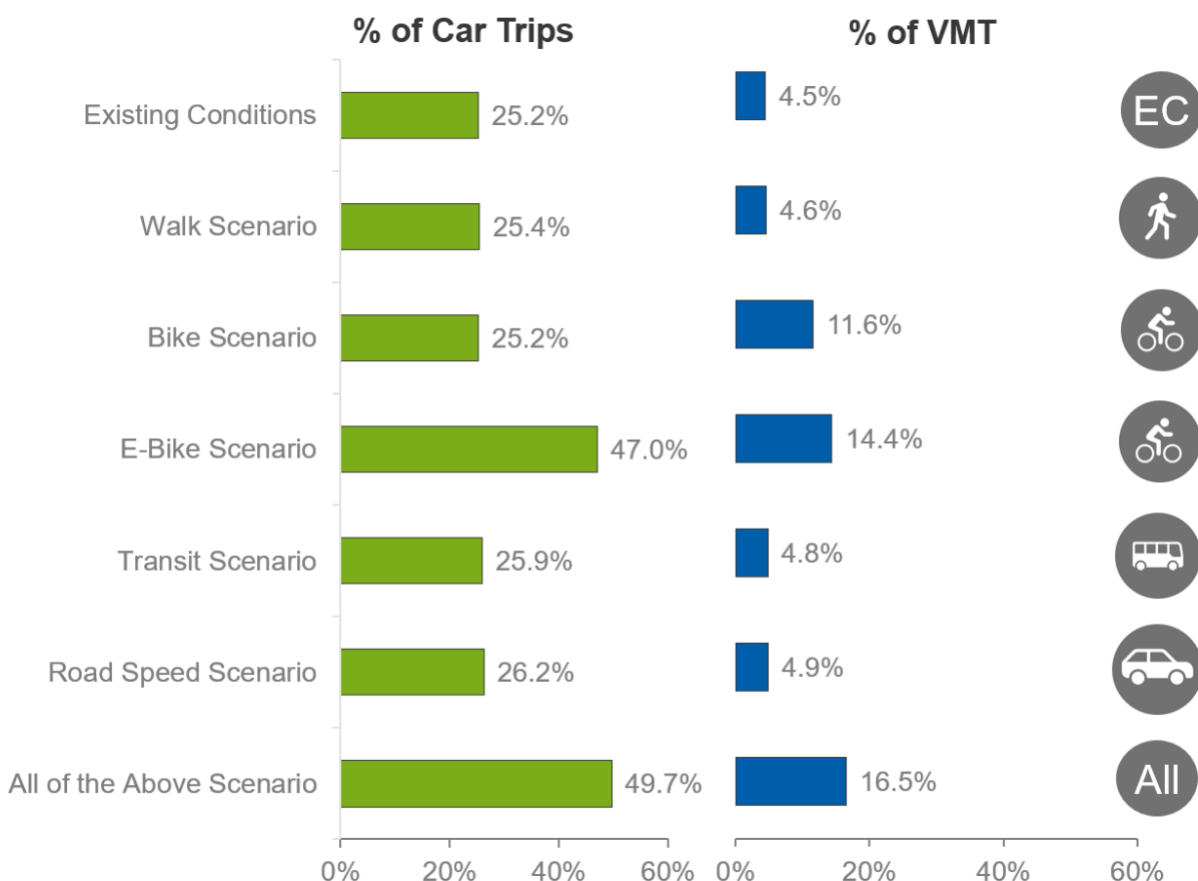


Figure 2 Percent of car trips and VMT for which the fastest feasible alternative is competitive

The all of the above scenario in Figure 2 shows that **if we made it safe and comfortable to walk and bike everywhere it is legal, made e-bikes available, doubled transit service, and reduced car speeds, half of car trips and 16.5% of VMT could competitively shift to another mode.** While the potential mode shifts in these scenarios are promising, it is important to remember that just because people *can* switch to another mode does not necessarily mean they *will* switch. The results also show that under existing conditions, people making 25% of car trips can switch to another mode but have not. More work is needed to assess how to most effectively prompt a shift to another mode once that mode is competitive. Overall, we can draw several conclusions from this analysis:

- Biking has the highest potential for mode shift. This potential can be further enhanced by building safe and comfortable bike infrastructure on all streets, but more work is needed to assess the extent to which this potential can be realized.
- Short trips have much higher mode shift potential. Compact land-use can provide a double-benefit of higher mode shift potential and shorter trips for those that remain in cars and is necessary to achieve greater VMT reduction.
- Even with broad changes to transportation infrastructure and service, many trips remain that cannot competitively shift. Vehicle electrification policies can be targeted towards travelers least able to switch modes.

The remainder of this report explores these findings in more detail.

Introduction

Achieving climate, livability, and safety goals requires significant shift away from driving to telework, transit, bicycling, and walking. As planning organizations begin to contemplate the scale of change needed to meet the climate crisis, many are turning towards target-based planning. Rather than predicting a most-likely future, policy makers instead set targets for a desirable future. For example, Minneapolis has set a target for 60% of trips within the city to occur by walk, bike or public transit by 2030 (City of Minneapolis, n.d.), St. Paul aims to reduce vehicle miles traveled (VMT) by 40% by 2040, and Minnesota has adopted a target to reduce per capita VMT by 20% by 2050 (Heggedal 2023).

Compared to the traditional “predict and provide” approach (Black 1990), target-based planning excels at providing a vision for the future that goes beyond the status quo (Lyons and Davidson 2016). However, target-based planning is likely to be most effective if the targets are both achievable and tied to the concrete actions necessary to meet them. This study aims to inform those discussions in the Twin Cities region by addressing three core questions:

1. With land use, transportation system, and travel patterns held constant, how much potential exists for travel to shift from driving towards other, less carbon-intensive modes?
2. To what extent does the mode shift potential vary by geography, demographics and trip types?
3. By how much may changes to the transportation system increase the mode shift potential?

We address these questions using a methodology grounded in observed data and transparent assumptions. Our approach is deliberately simple, and unlike travel forecasting models, we do not simulate changes to the population, or where people live, work, shop, and travel. Instead, we rely on 380,000 real-world trips observed in the 2019 and 2021 Travel Behavior Inventory (TBI) surveys. Our approach is not a replacement for travel models, but a complement to them. It aims to focus the discussion where it belongs—on possible strategies to achieve the targets—instead of on debates about modeling approaches and sensitivities. The results of this study can help guide infrastructure and services investment to support such behavioral changes while filling the gaps in the transportation system to make even greater shifts possible.

Methodology

The TBI surveys record the real-world travel behavior for a representative sample of households in the Twin Cities region in 2019 and 2021. Respondents are asked to install an app on their phone that records the locations they visit and prompts them to indicate the mode of each trip and the purpose of each stop. The resulting data indicate in detail where, when and how people in the Twin Cities region travel. Using these data, we observe both the trips made in cars, and those made by walking, biking or transit. For each observed car trip, we do the following:

First, we calculate the best **possible** walk path, bike path and transit path, considering the quality of the available infrastructure and transit service. The walk paths consider the presence of sidewalks and the functional class and speed limit of the adjacent road. The routes reflect that people prefer to walk on local roads with sidewalks than on high speed arterials with heavy traffic, and will go out of their way to avoid such less preferred. The bike routes consider a measure called Level of Traffic Stress (LTS) that reflects the availability of bike lanes and the discomfort of riding on busy, high-speed roads. The transit routes are based on published transit schedules and the actual departure/arrival time of the trip, with the routes calculated using a method similar to a smartphone mapping apps. Given this routing, we measure the travel of the preferred route for each mode and can compare that to the time for the car trip. We visually inspected a sample of routes and travel times for all modes to ensure they were logical and consistent with our knowledge of the region.

Our routing tools searched for paths that someone with options might use, taking into account safety, comfort and time.



Figure 3 Summary of routing assumptions by mode

Second, we consider which car trips could **feasibly** switch to another mode, given the possible routes. For example, it might be **possible** to walk 10 miles to work, but it is not necessarily **feasible** to do so. We determine the feasibility of shifting to another mode by introducing the “5% rule”:

If we have a reason to think that something is a barrier to shifting modes AND fewer than 5% of observed trips currently using that mode violate that barrier, then we consider doing so to be infeasible.

For walk trips, we think that walking is infeasible for long trips, and we observe that only 5% of current walk trips are longer than 1.6 miles. Therefore, we consider walking to be a feasible alternative only if the walking distance is less than 1.6 miles. This does not mean it is impossible walk further—some people do—but our interest is in shifting trips from car, and people who currently drive are likely less willing to walk long distances. Applying equivalent logic, bike trips are feasible if they are less than 10 miles in length, include less than 15% of their distance on high traffic-stress facilities, and there is no snow on the ground. Transit trips are feasible if both the origin and destination are within 0.9 miles of a transit stop, the waiting time is less than 36 minutes, and there are no more than 2 transfers. We considered other possible feasibility constraints, but found that they did not meet the

standard for exclusion of the 5% rule. These possible constraints considered included precipitation and temperature as a limit on walking and biking, as well as trip purpose for understanding whether people may be less willing to walk, bike or ride transit if they need to carry goods with them.

We also consider whether a trip can be completed within the available time without interfering with a work, school or escort passenger activity. For example, an afternoon trip between finishing work and picking up a child from daycare likely has limited flexibility on both ends, and there may not be sufficient time available to accommodate a longer travel time.

We examined existing walk, bike and transit trips, and defined a mode shift to be feasible if at least 5% use the mode under similar circumstances.

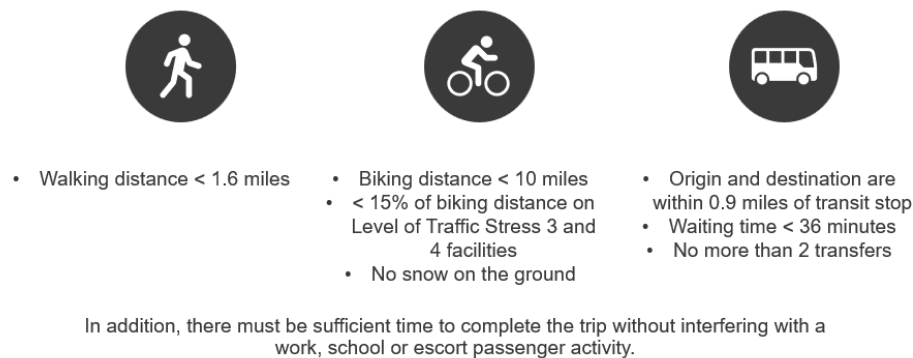


Figure 4 Summary of feasibility by mode

Third, we assess whether each feasible option is **competitive** by comparing the travel time for that alternative mode to the travel time for driving. The driving time considers traffic congestion using measures from a commercial speed data provider. The driving time also includes the time to walk between the car's parking location and the trip's origin/destination, which we assume varies from 1 minute in suburban locations to 4 minutes downtown. We plot the distribution of trips versus the travel time difference to understand their competitiveness, and in some cases the alternative mode is faster than driving due to congestion and parking time. For the purpose of summarizing the results, we consider a feasible option to be competitive if its travel time is no more than 15 minutes longer than the car trip it would replace. We also explore how the results differ for varying travel time thresholds.

Given these calculations, we summarize the data to provide insight in different dimensions to understand which modes are most competitive, where, and for whom. We apply this process again to test several scenarios to understand the extent to which changing the transportation system would result in more competitive travel by alternative modes. Additional details on the approach are included in the appendices.

Existing Mode Shift Potential

This section reports the potential for car trips to shift to walk, bike, transit, or any of these modes. It aims to address the question:

With land use, transportation system, and travel patterns held constant, how much potential exists for travel to shift from driving towards other, less carbon-intensive modes?

Potential shift to walk

To measure the possible walk routes, we create an “all streets” network from OpenStreetMap that includes the street system and off-street walking paths but excludes freeways and other facilities where walking is prohibited. We introduce a penalty to reflect that walkers tend to avoid roads with a higher walking level of traffic stress (LTS)—those roads with high speeds, high functional classes, or without sidewalks—using an approach developed by the Virginia DOT for project evaluation (Hardy, Bell, and McCahill 2019). The resulting paths reflect diversions from these LTS penalties, as shown in Figure 5, where someone may walk further to avoid a major arterial. We calculate the travel time along this longer route assuming a 3 mph walking speed.

Table 1 shows the feasibility constraints for walk, and the percentage of car trips and VMT that can feasibly switch according to each constraint. First, the data show that 22.8% of car trips have a walk alternative less than 1.6 miles long (compared to 95% of observed walking trips). Because these trips are by definition short, they comprise only 3% of VMT. Separate from the distance constraint, the data show that people have sufficient time in their schedule to complete 63% of car trips by walking instead, comprising 57% of VMT. In many cases, the schedule can accommodate these switches because the trips are the first or the last of the day, allowing travelers to leave earlier or arrive later at home. For a trip to be feasible, it must be feasible across all the constraints, and we find that overall, 18% of car trips and 2.2% of VMT could feasibly switch to walking.

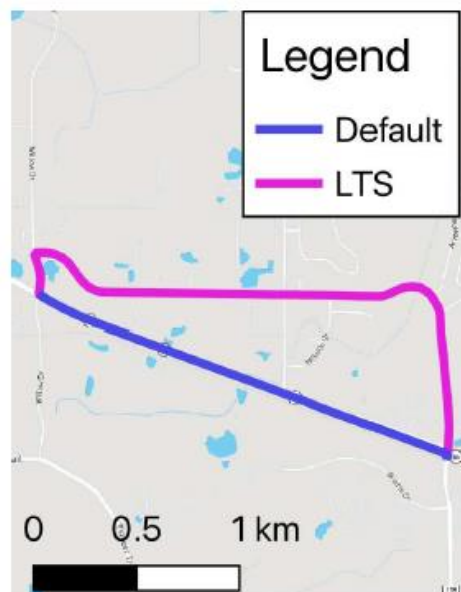


Figure 5 Example of walk trip with default (shortest distance) route versus LTS weighted route

Table 1 Percentage of car trips and VMT that can feasibly switch to walking

Feasibility Constraint	% of Car Trips	% of VMT
Walking distance < 1.6 miles	22.8%	3.0%
Sufficient time is available to complete the trip	63.0%	57.0%
Feasible across all constraints	18.0%	2.2%

Figure 6 illustrates the competitiveness of the walk as an alternative by plotting the frequency of trips versus the travel time difference. The call-out box below describes how to understand these plots.

In Figure 6, we observe a substantial frequency of trips in red where people currently walk. Most of these have travel times that are within 15 minutes of driving, and for many, the travel time is faster than driving. We also observe a sizable light blue area, indicating trips that could feasibly switch

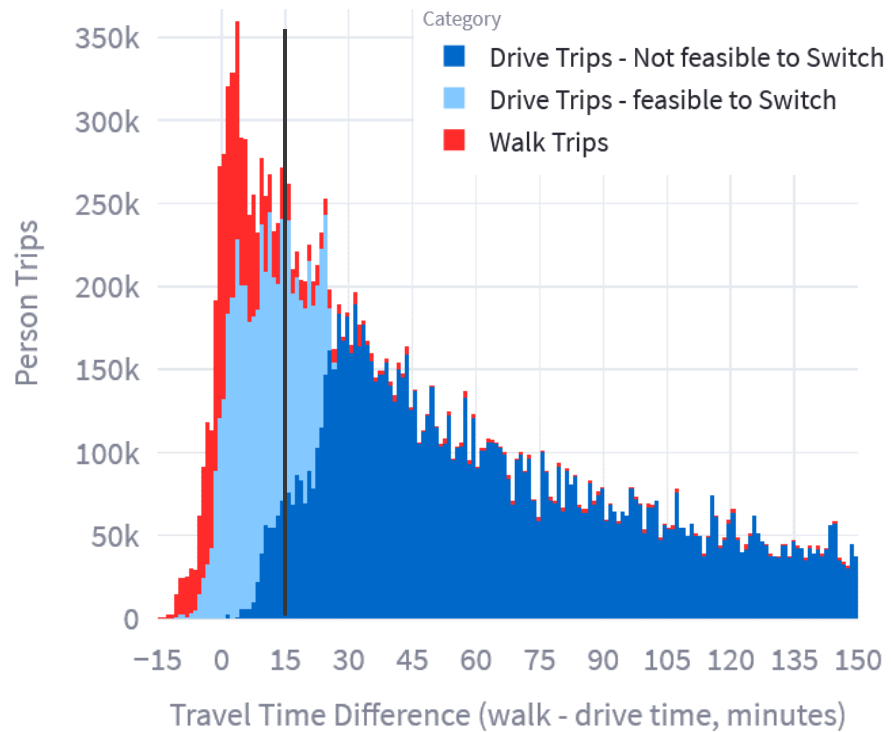


Figure 6 Travel time difference between walking and driving

Understanding travel time difference plots

Figure 6 and several subsequent figures illustrate the competitiveness each mode as an alternative to driving. In this call-out box, we explain how to understand these graphs, then focus on the findings in the text adjacent to each figure.

In these plots, the horizontal axis is the travel time difference between using the alternative mode and driving. Values greater than zero indicate that driving has a faster travel time. Values less than zero indicate that the alternative mode is faster. The vertical axis shows the number of trips for a specific difference in travel times. In general, we observe a higher frequency of trips clustered towards the left end of the graph, indicating that the travel times are competitive for both modes. In some instances, the travel time for the alternative mode is faster than driving, which can occur when cars experience traffic congestion, or when people must spend time walking from their parking place to their destination. Conversely, for some trips it might take 60 minutes or more longer to switch modes than to drive. In the case of Figure 6, these tend to be longer trips, where it might take very long time to walk 5 or 10 miles. The vertical black line shows a threshold of 15-minutes of travel time difference, and we consider trips to the left of this line to be “competitive”.

The color coding further segments the trips. The blue colors represent trips currently made by car, with the light blue indicating that those trips can feasibly switch to the alternative mode, and the dark blue indicating they cannot. For the dark blue areas, there may be some other barrier to changing modes, even though the travel time may be competitive. The red areas represent trips currently made by the alternative mode. In the example of Figure 6, most people who currently walk have travel times that are competitive with driving, but some choose to walk even though it takes longer—maybe they enjoy the walk or maybe they do not have a car.

to walking. Most of these feasible trips have walking times that are competitive with driving. Unfortunately, the largest area under the curve is dark blue, indicating drive trips where it is not feasible to switch to walking. Most of these infeasible trips are too long to reasonably accommodate walking.

Table 2 summarizes these same data for three specific thresholds of travel time difference. Among trips currently made by car, 5.5% can feasibly switch to walking with a travel time within 5 minutes of driving, 13% can feasibly switch with a travel time within 15 minutes, and 18% can feasibly switch with a travel time within 30 minutes. If those trips did switch to walking, it would reduce VMT by 0.4% for the 5-minute threshold, 1.2% for the 15 minute threshold and 2.2% for the 30 minute threshold.

Table 2 Percentage of car trips and VMT that can feasibly switch to walking within specific time differences

Travel time difference (walk time – drive time)	% of Car Trips	% of VMT
Walking is within 5 minutes of driving	5.5%	0.4%
Walking is within 15 minutes of driving	13.0%	1.2%
Walking is within 30 minutes of driving	18.0%	2.2%

Potential shift to bike

As with walking, we calculated bike routes using an “all streets” network derived from OpenStreetMap, but filtered to include all bicycle facilities and exclude roads where bicycling is prohibited. Previous research indicates that bicyclists are highly averse to riding on busy and high-speed streets, especially without bike lanes (Hood et al. 2014; Broach 2016). To reflect this aversion, our routing considers the bicycle level of traffic stress (LTS) (Furth, Mekuria, and Nixon 2016), as illustrated in Figure 7 and previously used by the

University of Minnesota (Owen and Murphy 2020). We calculate the bike routes by imposing a 10% penalty for biking on LTS 2 facilities and assuming that bicyclists who encounter LTS 3 and 4 facilities will dismount and walk their bikes until they reach a more comfortable street or path. While existing cyclists do sometimes ride on these facilities, a mode shift suggests the need to attract new cyclists who will be less confident. We also assume bicyclists will avoid hills if it does not make their trip too much longer. These penalties result in routes that avoid busier facilities, as the example in Figure 8 shows. We assume that bicyclists travel at 12 mph along their preferred route and that it takes 1 minute at either end to lock or unlock a bike.

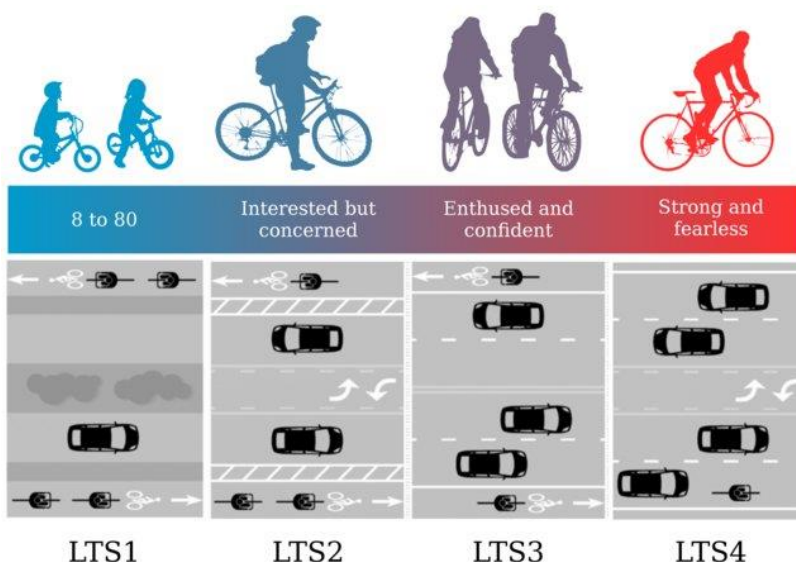


Figure 7 Bicycle level of traffic stress (LTS) illustration



Figure 8 Example of bike trip with default (shortest distance) route versus LTS weighted route

Table 3 shows the feasibility constraints for bike, and the percentage of car trips and VMT that can feasibly switch according to each constraint. Among observed bike trips, only 5% are longer than 10 miles in length and 67% of car trips have a preferred bicycle route within this distance, amounting to 27% of VMT. The TBI data were collected year-round, allowing us to analyze weather conditions. We found that fewer than 5% of bicycle trips are made when snow is on the ground, and 76% of car trips and VMT are made when there is not snow on the ground. 87% of car trips and 94% of VMT have a preferred bicycle route alternative with no more than 15% of the distance on LTS 3 or 4 facilities. People’s travel schedules have sufficient time to complete 71% of car trips and 64% of VMT by bike instead. When considering all four constraints, biking is a feasible option for 33% of car trips and 13% of VMT.

Table 3 Percentage of car trips and VMT that can feasibly switch to biking

Feasibility Constraint	% of Car Trips	% of VMT
Bicycle distance is < 10 miles	67.3%	26.9%
There is not snow on the ground	76.6%	76.4%
No more than 15% of the distance is on LTS 3 or 4 facilities	86.7%	93.9%
Sufficient time is available to complete the trip	71.4%	63.7%
Feasible across all constraints	32.5%	13.0%

Figure 9 shows the distribution of travel time differences between biking and driving in the same format as used previously. The call-out box on Page 9 describes how to interpret these travel time difference plots. Compared to walking, a greater number trips have smaller travel time differences because biking is faster. However, a substantial portion of trips that would otherwise be competitive do not meet the feasibility constraints for bicycling. The figure also includes less red, indicating fewer observed biking trips, even when biking is both feasible and competitive.

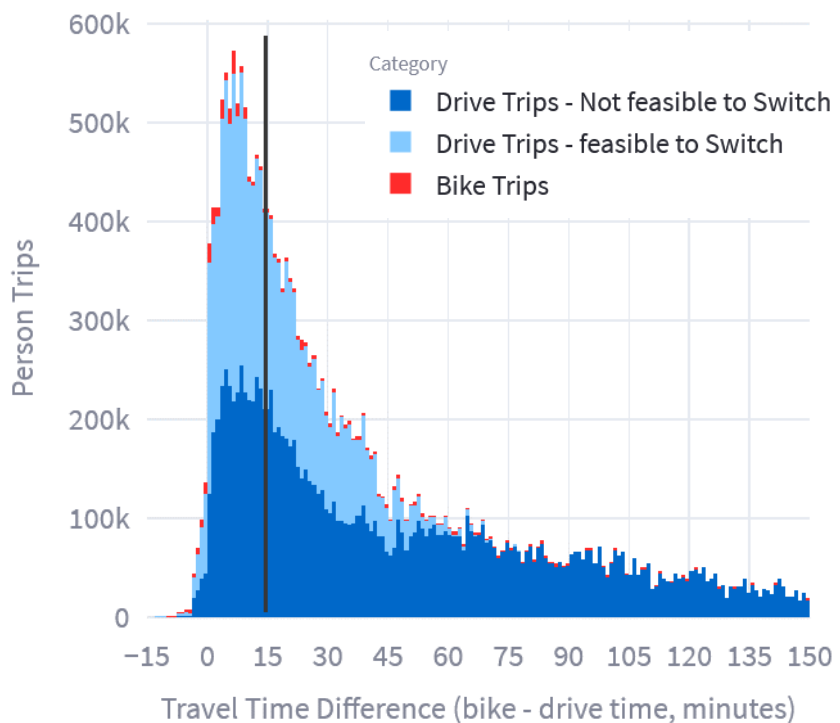


Figure 9 Travel time difference between biking and driving

Table 4 summarizes these results for specific time thresholds. Among car trips, 6%, 17% and 26% could feasibly switch to biking within 5, 15 and 30 minutes of the driving time, respectively. If all of these trips did switch, VMT would be reduced by 0.8%, 3.4% and 7.7% for the respective thresholds.

Table 4 Percentage of car trips and VMT that can feasibly switch to biking within specific time differences

Travel time difference (bike time – drive time)	% of Car Trips	% of VMT
Biking is within 5 minutes of driving	5.9%	0.8%
Biking is within 15 minutes of driving	16.7%	3.4%
Biking is within 30 minutes of driving	25.6%	7.7%

Potential shift to transit

We calculated using a method similar to a smartphone's mapping app's method. The method uses data from the General Transit Feed Specification (GTFS) files published by Metro Transit and Minnesota Valley Transit Authority for October 2019. GTFS is a standard format operators use to publish their schedules for use in mapping apps. Our routing method considers the exact departure or arrival time of the trip and reflects the scheduled transit vehicles and possible transfers available at that time, as well as the walking time to, from or between transit vehicles. For trips to work, school, or medical appointments, we assume the arrival time is fixed and calculate the latest departure from the trip's origin that would reach the destination in time. For other trips, we calculate the earliest arrival given the departure time. Because the waiting time and route vary based on the trip's timing, we allowed arrivals up to 5 minutes late at work, school, or medical appointments, and departures up to 5 minutes early for other trips, if they made the trip at least 15 minutes faster. Figure 10 shows a transit path in Minneapolis.

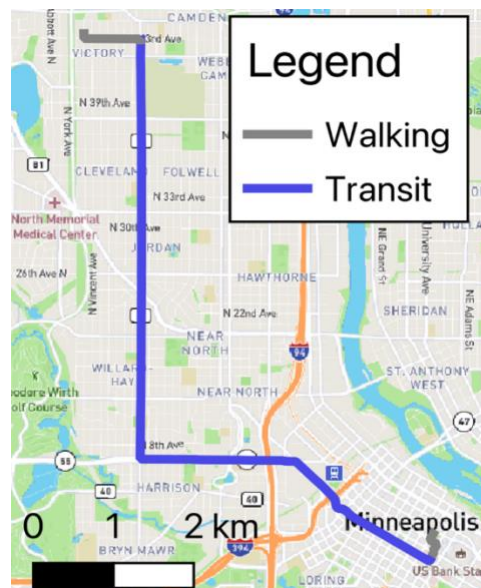


Figure 10 Example of a trip's transit path, including walk and in-vehicle segments

Table 5 summarizes the feasibility constraints for transit and the percentage of car trips and VMT that meet each. A valid transit path with 2 or fewer transfers is possible for 61% of car trips, corresponding to 48% of VMT. Note that we impose the transfer limit (and a loose 1.5 mile limit on walking distance) in the routing method itself to avoid creating a large number of complex and infeasible paths, so the subsequent constraints rely on first meeting this constraint. Walking distance to or from a transit vehicle is also a barrier to using transit and reflects the overall coverage of the bus system. For 48% of car trips and 37% of VMT, the possible transit path involves walking less than 0.97 miles from the origin to the first transit stop, and for about the same percentages, the preferred transit path involves walking less than 0.93 miles from the final transit stop to the destination. For 45% of car trips and 31% of VMT, the possible transit path involves waiting less than 64 minutes for the first vehicle, implying that a route running 30 minute or better headways would not be constraining. Each of these constraints is derived from the 5% rule, with fewer than 5% of existing transit trips violating them.

Table 5 Percentage of car trips and VMT that can feasibly switch to transit

Feasibility Constraint	% of Car Trips	% of VMT
Valid transit path found with 2 or fewer transfers	61.4%	47.5%
Walk access distance < 0.97 miles	48.1%	37.0%
Walk egress distance < 0.93 miles	48.4%	36.7%
Waiting time for first vehicle < 64 minutes	44.4%	30.5%
Sufficient time is available to complete the trip	34.6%	25.8%
Feasible across all constraints	19.7%	12.4%

Sufficient time is available to complete the trip by transit for 35% of car trips and 26% of VMT. Considering all constraints together, 20% of car trips and 12% of VMT could feasibly be replaced by transit.

Figure 11 shows the travel time difference between transit and driving. The histogram shows a substantial share of feasible trips with relatively large travel time differences, but also a substantial share of current transit trips with a relatively large travel time difference. The observed trips highlight that people use transit for reasons beyond travel time—maybe they do not have a car or cannot drive—highlighting its value as an essential service, but also the challenges of adding new riders.

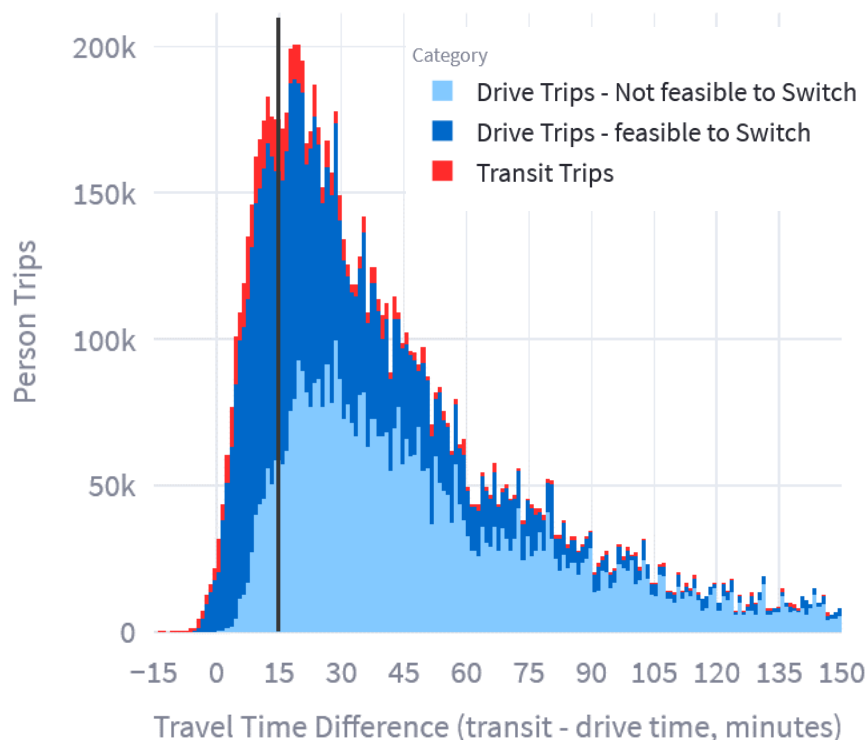


Figure 11 Travel time difference between transit and driving

Table 6 summarizes the percentage of car trips and VMT that can feasibly switch to transit within 5, 15 and 30 minutes of driving. Overall, 1.4% of car trips and 0.2% of VMT can feasibly switch to transit and experience travel times within 5 minutes of driving, 6.0% of car trips and 1.2% of VMT can switch to transit with travel times within 15 minutes of driving, and 12% of car trips and 3.7% of VMT can switch to transit with travel times within 30 minutes of driving.

Table 6 Percentage of car trips and VMT that can feasibly switch to transit within specific time differences

Travel time difference (bike time – drive time)	% of Car Trips	% of VMT
Transit is within 5 minutes of driving	1.4%	0.2%
Transit is within 15 minutes of driving	6.0%	1.2%
Transit is within 30 minutes of driving	11.8%	3.7%

Potential shifts to any mode

Some car trips can feasibly shift to only one alternative mode, while others can feasibly shift to multiple options. The middle columns of Table 7 summarize these feasible shifts by mode (as previously reported in Tables 2, 4 and 6) and also shows that 48% of car trips and 22% of VMT can feasibly switch to at least one non-car mode. For those feasible shifts, the rightmost columns in Table 7 show which mode is fastest. Walk is the fastest alternative for 9% of car trips, bike is fastest for 30% of car trips, and transit is fastest for 9% of car trips. As a percentage of VMT, the values are 1%, 12% and 9% for walk, bike and transit, respectively. There is no feasible alternative to driving for over half of car trips and over three-fourths of VMT.

Table 7 Percentage of car trips and VMT that can feasibly switch to each mode, and which is the fastest

Feasibility Constraint	Feasible to shift to mode		Mode is the fastest alternative	
	% of Car Trips	% of VMT	% of Car Trips	% of VMT
Walk	18.0%	2.2%	9.3%	1.0%
Bike	32.5%	13.0%	29.5%	11.7%
Transit	19.7%	12.4%	9.4%	9.0%
Any non-car mode	48.1%	21.7%	48.1%	21.7%

These results show that even though walk and transit are the fastest alternative to driving for a similar percentage of car trips, transit is the fastest alternative for a much higher percentage of VMT because transit can serve longer trips. Because the transit wait times vary by departure and arrival times, the fastest alternative may vary by time-of-day even for the same origin and destination. The fastest alternative also varies by season because we consider biking infeasible when snow is on the ground. For these reasons, many travelers may find value in choosing among multiple feasible alternatives.

Figure 12 shows a histogram of the travel time difference between the fastest alternative mode and driving, color coded by which mode is fastest. The large green area suggests that biking is the fastest alternative for the largest portion of car trips. The gray area indicates that where walking is the fastest alternative, it is usually within 15 minutes of driving. The light blue area indicates that where transit

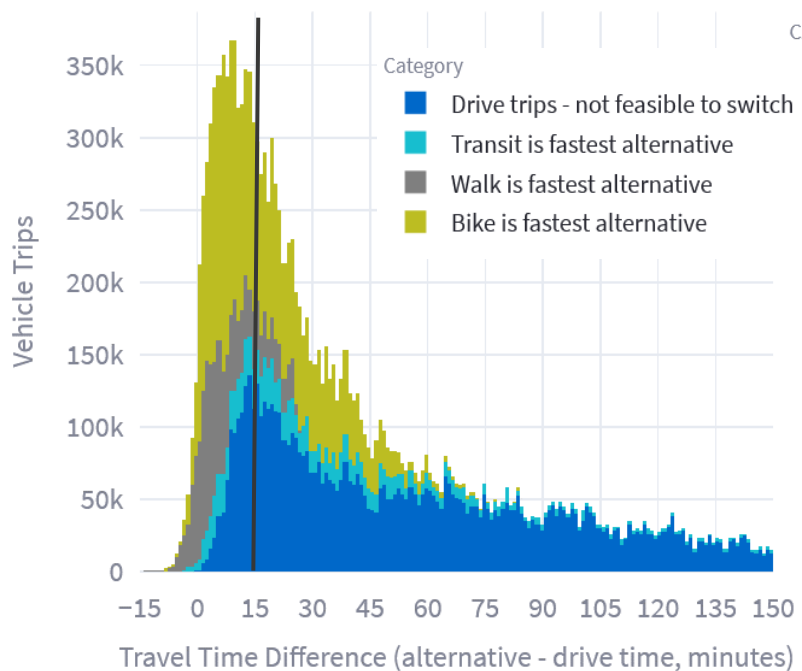


Figure 12 Travel time difference between fastest non-car mode and driving

is the fastest alternative, the travel time difference with driving varies widely, suggesting that it may often be the only feasible alternative. The dark blue areas are trips that cannot feasibly shift from car to another mode—even though some alternatives would otherwise be competitive, there constraints prevent them from feasibly shifting.

Table 8 shows the percentage of car trips and VMT that can feasibly shift to any non-car mode within the specified travel time thresholds. Approximately 10% of trips and 1% of VMT can switch to another mode within 5 minutes of driving, 25% of trips and 4.5% of VMT can switch within 15 minutes and 37% of trips and 10% of VMT can switch within 30 minutes.

Table 8 Percentage of car trips and VMT that can feasibly switch to any non-car mode within specific time differences

Travel time difference (alternative time – drive time)	% of Car Trips	% of VMT
Any feasible option is within 5 minutes of driving	10.2%	1.1%
Any feasible option is within 15 minutes of driving	25.2%	4.5%
Any feasible option is within 30 minutes of driving	37.2%	10.2%

Variation by geography, demographics and trip purpose

Here, we explore the question:

To what extent does the mode shift potential vary by geography, demographics and trip types?

First, we examine the variation by geography, as Figure 13 shows. The left portion of the figure shows the percentage of car trips that can feasibly switch to another mode, and the right portion shows the percentage that can competitively switch—that it is both feasible and within 15 minutes of the driving time. Both portions show that the percentage of trips that can switch is higher in the urban areas than it is in the suburban or rural parts of the region. However, many of the smaller town centers, such as Shakopee and Hastings have high mode shift potential. The mode shift potential in the urban areas and town centers is higher because the destinations are closer allowing for shorter trips, and because the transportation infrastructure is better able to accommodate non-car trips.

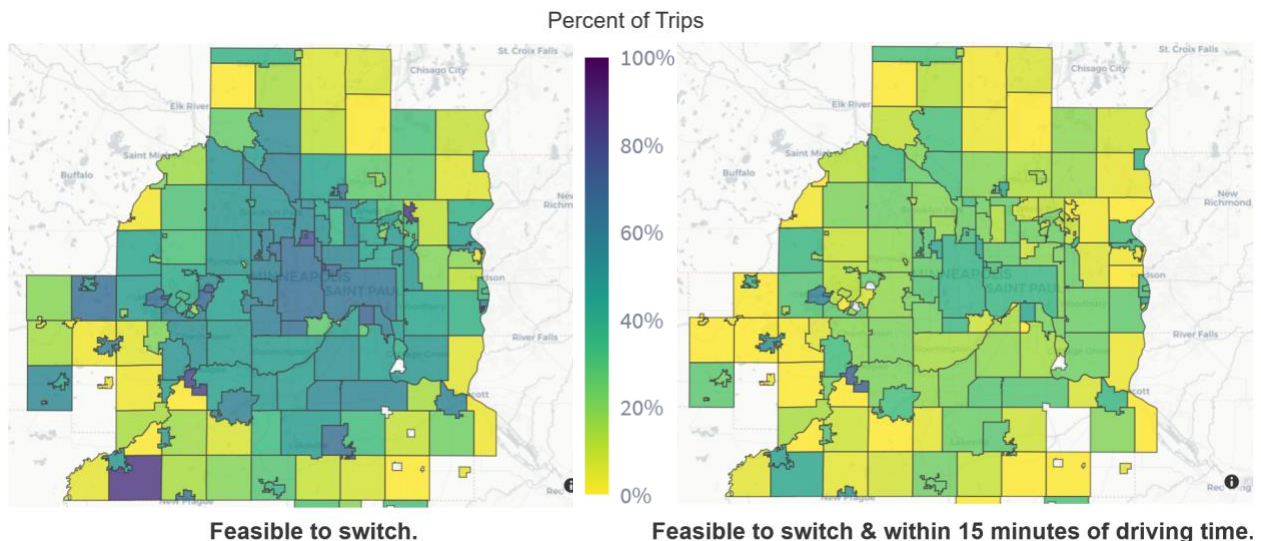


Figure 13 Percentage of car trips that can feasibly and competitively shift to another mode

Figure 14 shows the percentage of cars trips that can feasibly shift to another mode, segmented by the household income of the trip maker. In general, lower income travelers are most able to shift, but also make the fewer trips. This difference raises questions about how to equitably share the responsibility for reducing VMT.

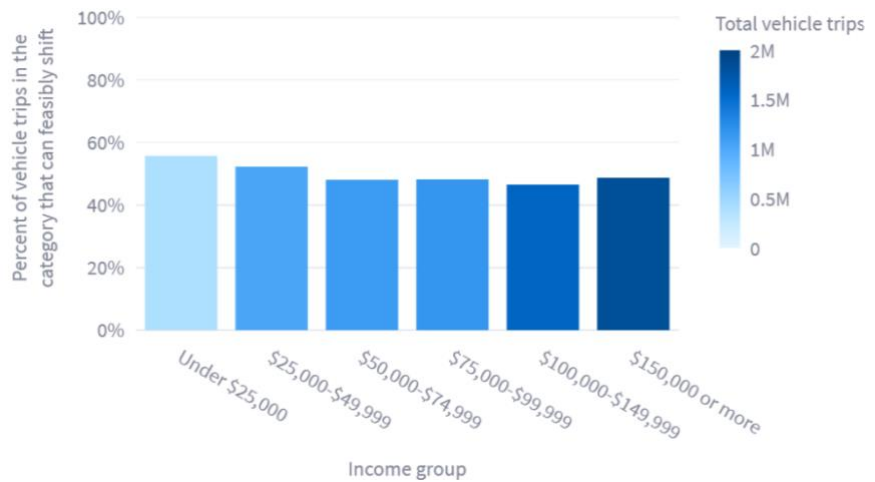


Figure 14 Percentage of car trips by income group that can feasibly shift to another mode

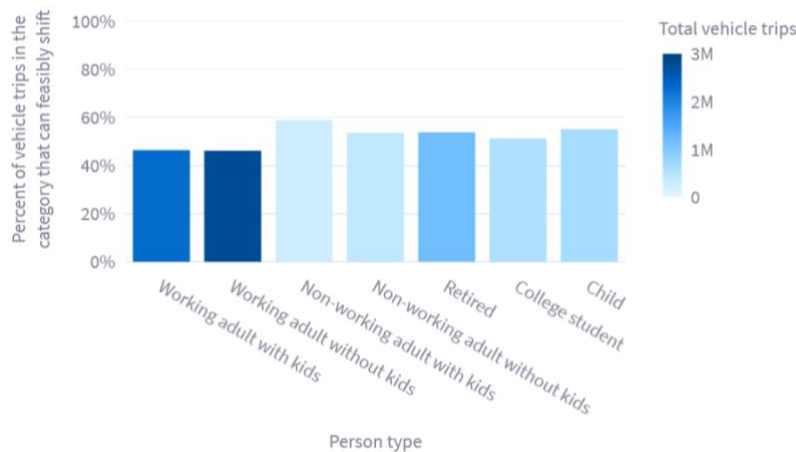


Figure 15 Percentage of car trips by person type that can feasibly shift to another mode

Figure 14 shows the percentage of car trips by person type that can feasibly shift to another mode. Working adults are least able to shift their travel to other modes due primarily to the time constraints imposed by work. Working adults also make the most trips, as indicated by the darker shades of blue. Non-working adults are most able to shift to other modes, but it is important to note that while our data do reflect the constraints of childcare and school pickups and drop-offs, they do not fully reflect the challenges of traveling with children.

Figure 16 shows the percentage of car trips by gender of the trip-maker that can feasibly shift to another mode. A slightly lower share of car trips made by men can feasibly shift, presumably due to differences in where and when the trips are made.

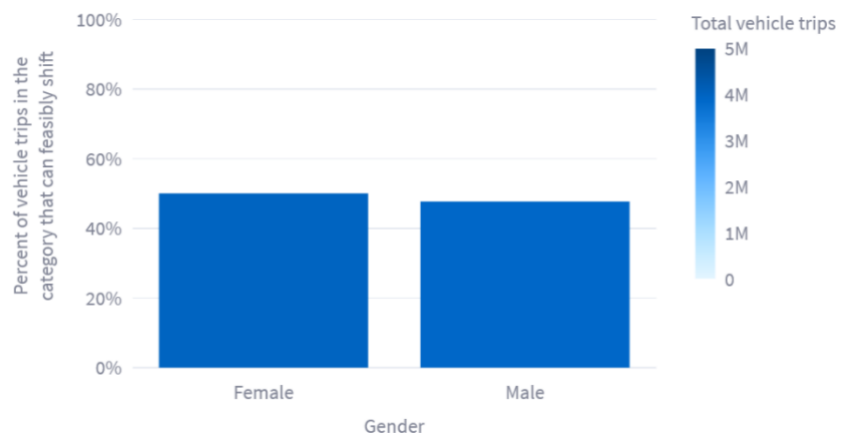


Figure 16 Percentage of car trips by gender that can feasibly shift to another mode

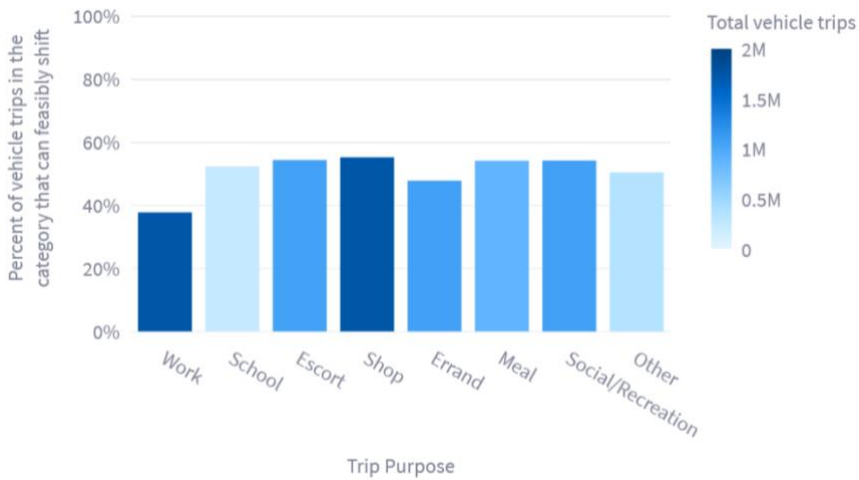


Figure 17 Percentage of car trips by trip purpose that can feasibly switch to another mode

Figure 17 shows the percentage of car trips by purpose that can feasibly shift to another mode. Trips to and from work are the least able to shift due to the time constraints of work and the longer distances involved. Shopping trips are most able to shift, because the trips tend to be shorter, and retail areas are often close to residential areas.

We did not explore the differences by race and ethnicity due to inconsistent recording of these measures.

Possible Actions to Increase Mode Shift Potential

Above, we examine how many car trips and how much VMT could feasibly and completely shift to walk, bike or transit assuming land use, the transportation system and travel patterns remain as currently observed. Next, we build upon that analysis to consider:

By how much may changes to the transportation system increase the mode shift potential?

To provide a starting point for discussions about how to effectively increase the mode shift potential, we evaluated the potential of six very broad scenarios to increase the competitiveness of alternative modes:

- Walk infrastructure: it is safe and comfortable to walk everywhere it is legal to do so;
- Bike infrastructure: it is safe and comfortable to bike everywhere it is legal to do so;
- E-bikes: additionally, everyone has an e-bike;
- Transit frequency: transit frequency is doubled for all routes;
- Roadway speed: speed limits are reduced based on NACTO guidelines; and
- All of the above.

We describe each in further detail on the following pages.

Improving walk infrastructure

The street environment affects people's comfort and safety when walking. Most people would rather walk along a sidewalk on a quiet street than along a busy road with traffic speeding past. In this study, we reflect those preferences by selecting walking routes that are weighted to avoid higher class and higher speed streets. This scenario imagines a region where every street is safe and comfortable to walk along. This ideal environment includes sidewalks everywhere, speed limits no more than 30 mph, and roads no wider than four lanes. People would walk the shortest distance to their destination, saving travel time.

Figure 18 shows the results of this scenario, compared to the existing conditions. It focuses on competitive shifts, where a competitive shift must be feasible and provide a travel time close to driving. Specifically, Figure 18 shows the percentage of car trips that can competitively shift to walking within 5, 15 or 30 minutes of the travel time of driving. It also shows the percentage of VMT that these car trips comprise. The results show that improving walk infrastructure would increase the competitiveness of walking slightly.

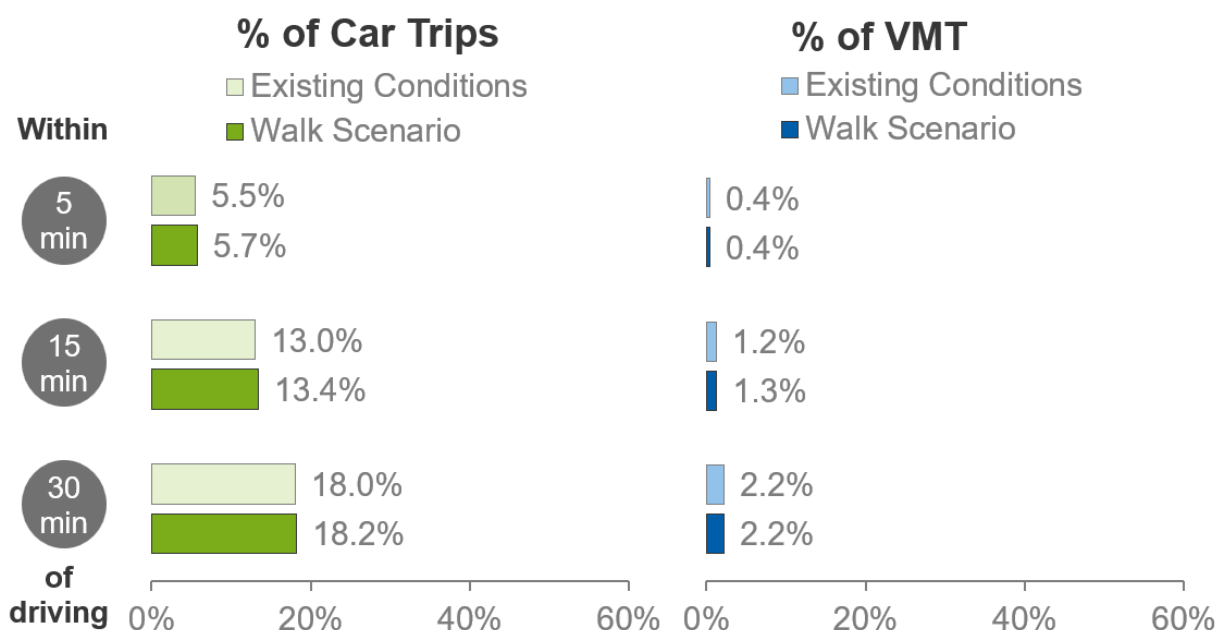


Figure 18 Percentage of car trips and VMT that can competitively shift to walking for a scenario where walking is safe and comfortable on all streets

Improving bike infrastructure

Many bicyclists are reluctant to ride on busy roads due to safety concerns and the stress of being near traffic. We reflect this reluctance by selecting bike routes that avoid roads with a high Level of Traffic Stress (LTS) (see Figure 7). This scenario imagines a region where everyone can safely and comfortably bike on any street. We represent this scenario by coding all streets except freeways with LTS 1, reflecting the best possible biking conditions. Bicyclists would therefore save travel time on each trip, because they could take the most direct route to their destination, without diverting or dismounting to avoid busy streets.

Figure 19 shows the results of this scenario, compared to existing conditions. Under this scenario, many more car trips could feasibly shift to biking and maintain competitive travel times, across all three time thresholds. If we consider a 15-minute travel time difference to be competitive, more than twice as many car trips could shift to bike (38.9% vs 16.7%) and more than three times as much VMT could shift to bike (11.0% vs 3.4%). Biking holds a high potential as an alternative to driving, but the quality of the existing bike infrastructure in the Twin Cities region is a major barrier to achieving that potential.

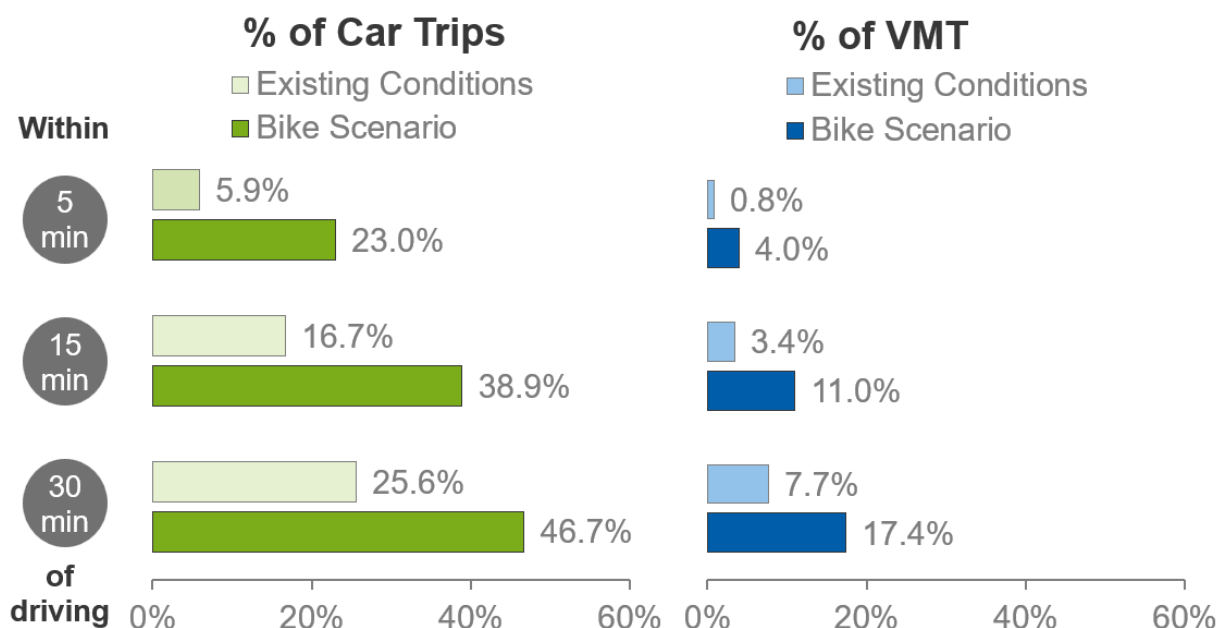


Figure 19 Percentage of car trips and VMT that can competitively shift to biking for a scenario where biking is safe and comfortable on all streets

Universal e-bikes

In the past few years, e-bikes have become more widely available and popular. The battery-electric motor on an e-bike reduces the effort required to pedal, improves acceleration, and increases the average speed. It therefore makes biking easier and faster. In this scenario, we build upon the bike scenario described above and imagine that in addition to the bike infrastructure improvements, everyone has access to an e-bike. We represent this change by increasing the biking speed by 13%, and removing avoidance of steep hills.

Figure 20 shows the results. Compared to the bike scenario, e-biking is more competitive with driving: 43% of car trips and 14% of VMT could feasibly shift to e-biking within 15 minutes of the driving time.

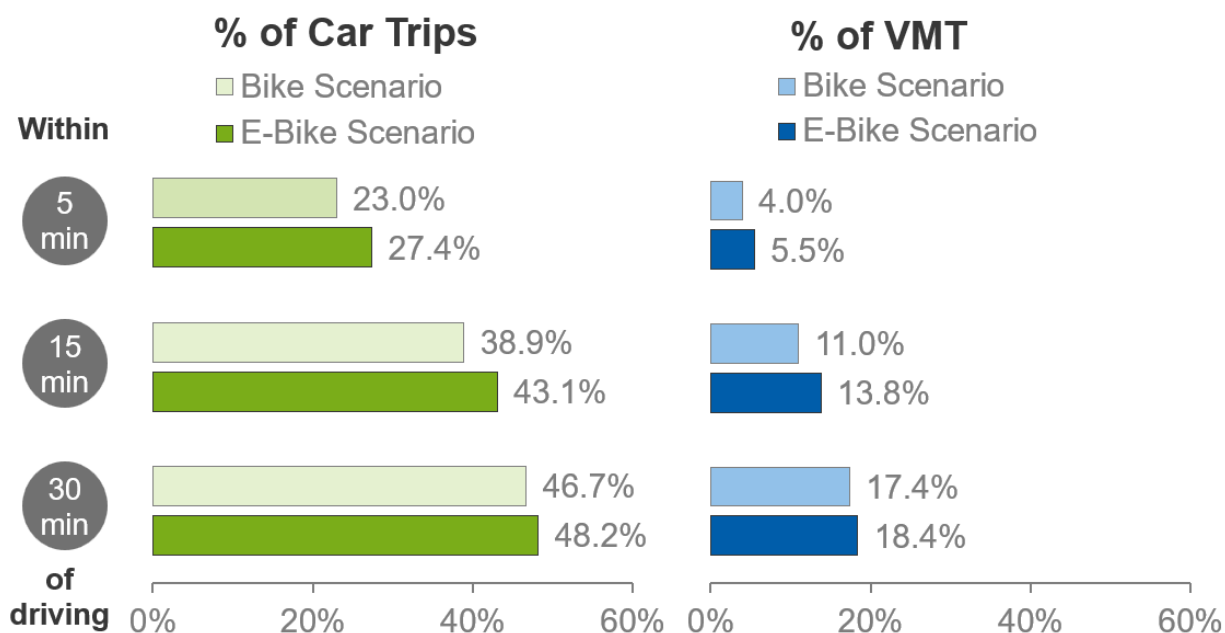


Figure 20 Percentage of car trips and VMT that can competitively shift to biking for a scenario where biking is safe and comfortable on all streets and everyone has access to an e-bike

Double transit frequency

Traveling by public transit incurs travel time in several parts. The traveler must walk to the stop, wait for the bus or train, ride to another stop, and walk to their destination. Trips that require a transfer involve another walking and waiting step. Therefore, there are several ways to improve the competitiveness of transit: by adding stops closer to the desired origins and destinations (increasing coverage), by increasing the speed of the transit vehicles (such as through exclusive bus lanes or limited-stop service), or by reducing waiting times through increased service frequency. In this scenario, we focus on the latter, and imagine a region in which the same transit routes are operated at double their current frequency. Thus, a route that currently runs twice per hour would instead run four times per hour, reducing the average headway from 30 minutes to 15 minutes, saving travelers up to 15 minutes of waiting time, depending on when they depart. Travelers may also select different paths, particularly if the shorter waiting times open more opportunities for transfers.

Figure 21 shows the effect of doubling transit frequencies. With this improved transit service, 8.0% of car trips could be completed by transit within 15 minutes of the driving time, compared to 5.8% currently. With improved transit service, 1.8% of VMT could be shifted to transit within 15 minutes with driving time, compared to 1.2% currently.

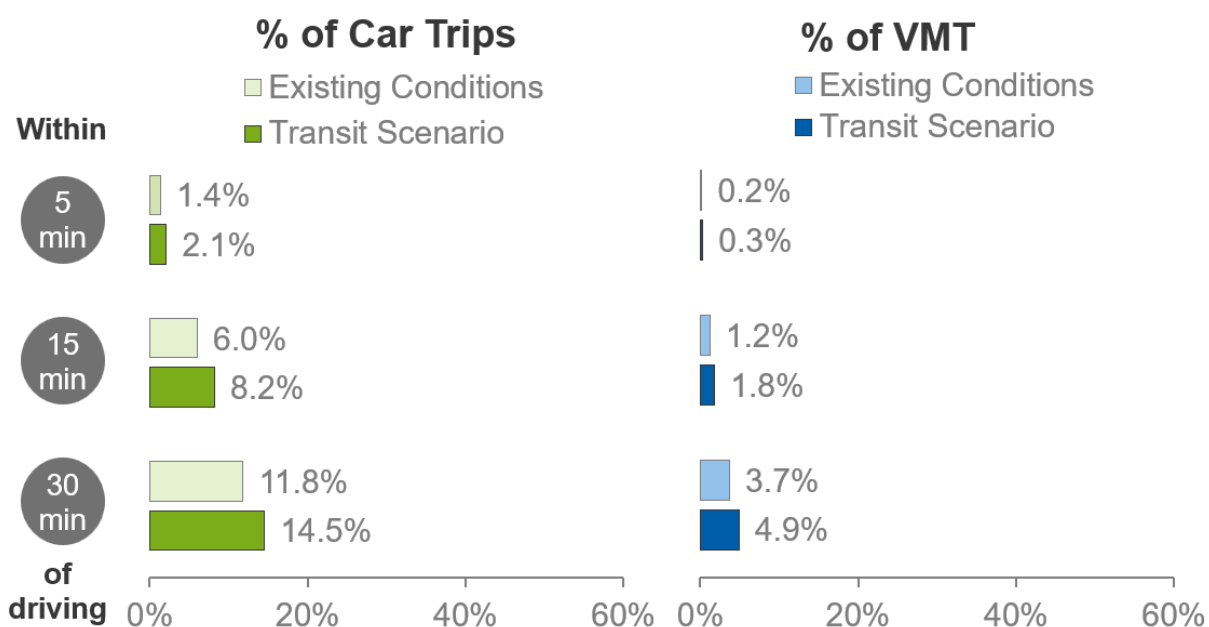


Figure 21 Percentage of car trips and VMT that can competitively shift to transit for a scenario where transit frequency is doubled on all routes

Reduce roadway speeds

The competitiveness of more sustainable modes depends both on the feasibility and travel times of those modes, and on the travel time to drive. Therefore, it is also possible to increase the competitiveness of other modes by reducing driving speeds with lower speed limits and/or street designs that naturally limit the driving speed. Here, we consider a scenario where the travel speeds on all streets in the Twin Cities region are adjusted to be consistent with guidance from the National Association of City Transportation Officials (NACTO). As a complement to the American Association of State Highway and Transportation Officials' (AASHTO's) "green book", NACTO publishes street design guidance focused on serving the mix of car, walk, bike and transit users found in urban and suburban areas (NACTO 2013). Their current guidance for urban areas (NACTO 2020) suggests:

- 10 mph on shared streets and alleys,
- 20 mph on minor streets,
- 25 mph on major streets, with a variation of up to 25 mph for streets in low density areas with well-protected places for people to walk and bike.

We apply the 10 mph limit to shared streets and alleys and the 20 mph limit to locals and collectors. In urban areas, we apply a 25 mph speed limit to arterials, and 25 mph in suburban areas. In addition, we limit freeway speeds to 55 mph in urban areas and 70 mph in suburban and rural areas. We represent these changes in a scenario by limiting travel speeds on those types of facilities to no more than the values indicated, assuming the streets are designed such that drivers comply. In both the base case and the scenario, we set the free flow travel speed (excluding congestion) to 80% of the speed limit to account acceleration and deceleration time from intersections.

Figure 22 shows the results of this road speed scenario, compared to existing conditions. It shows that these changes would slightly increase the competitiveness of switching to any non-car mode. There are two ways to view the modest changes resulting from this scenario: that these specific road speed changes provide limited benefit in terms of increasing the mode shift potential, or that they can be implemented with limited cost to those who currently drive. Further, these road speed changes would provide other benefits. First, they would improve the walking and biking conditions and would therefore likely be necessary to implement the walk and bike scenarios described above. Second, they would have safety benefits, with research showing that reducing roadway speeds can reduce both total crashes and fatal and serious injury crashes (NACTO 2020).

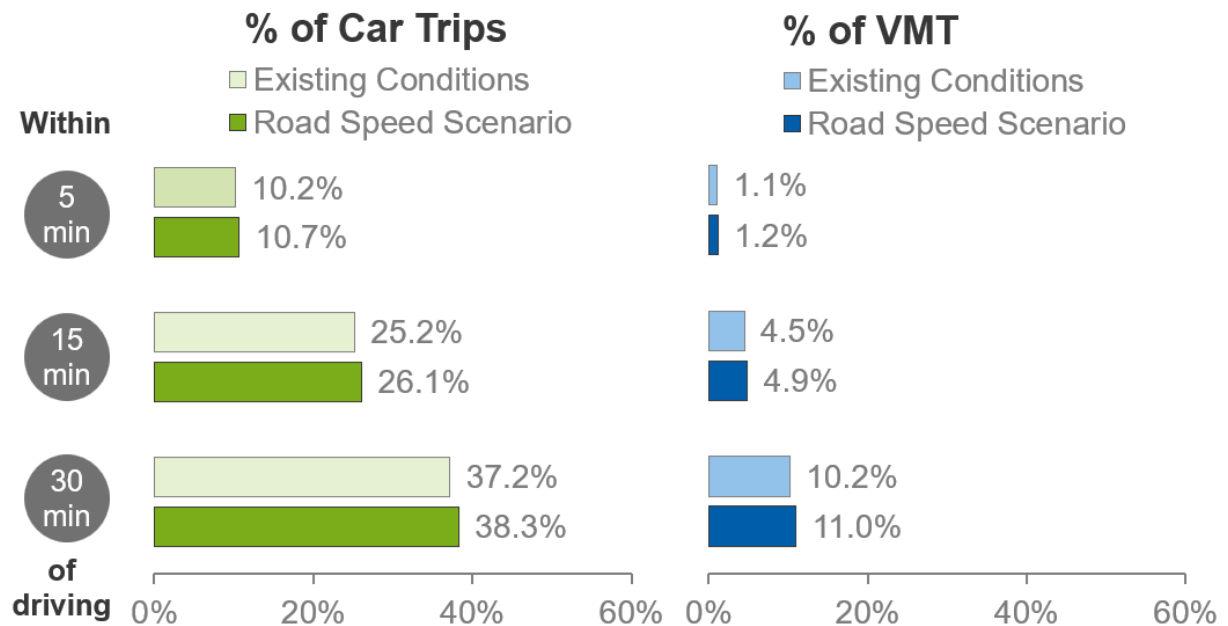


Figure 22 Percentage of car trips and VMT that can competitively shift to walking, biking or transit for a scenario where roadway speeds are limited based on NACTO guidelines

All of the above

More than one of the above scenarios could be implemented together, and they may be complementary. Limiting roadway speeds would also improve the walk and bike environment of many streets, and improvements to the walk and bike environment may be mutually beneficial. Here, we envision an all of the above scenario, where every street is safe and comfortable for both walking and biking, everyone has access to an e-bike, transit frequency is doubled throughout the region, and roadway speeds are limited consistent with NACTO guidance.

Figure 23 shows the mode shift potential for this scenario, compared to existing conditions. It shows that the percentage of car trips that can feasibly shift to walk, bike or transit within 15 minutes of driving doubles, and the percentage of VMT that can feasibility shift within 15 minutes of driving more than triples.

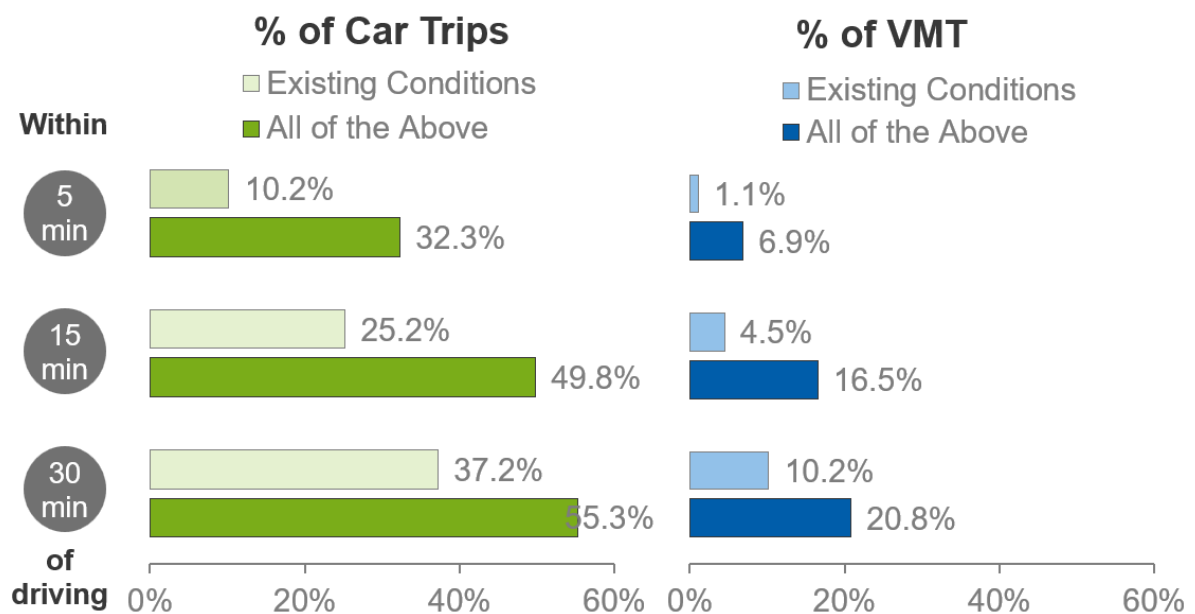


Figure 23 Percentage of car trips and VMT that can competitively shift to walking, biking or transit for a scenario where all of the above improvements are implemented

Scenario comparison

Figure 24 compares the results of each scenario, focusing on feasible shifts from car to any other mode that has a travel time within 15 minutes of driving. All scenarios would increase the competitiveness of non-car modes, with the bike scenario providing the largest individual effect, and the all of the above scenario providing the largest overall effect.

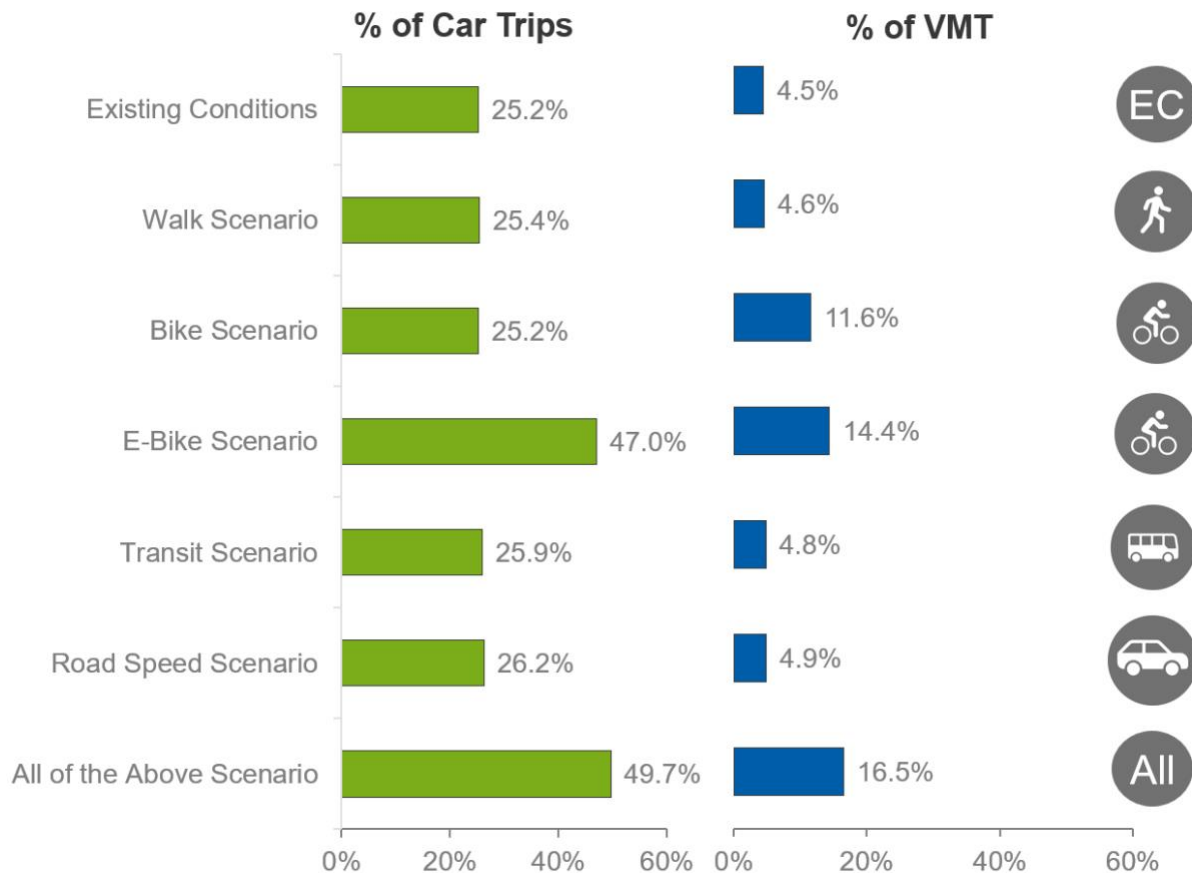


Figure 24 Percentage of car trips and VMT that can feasibly shift to walking, biking or transit with travel times within 15 minutes of driving for transportation improvement scenarios

Conclusions

In this study, we consider the potential to reduce vehicle miles traveled (VMT) in the Twin Cities region that might be achieved by incentivizing people to switch from traveling in cars to walking, biking or riding transit. We do so by estimating the maximum feasible shift of car trips to alternative modes given existing land-use and travel patterns, and whether those shifts are competitive (within 15 minutes of the driving time). **Overall, in the Twin Cities region today, 25% of car trips could shift to walk, bike or transit with travel times within 15 minutes of driving, and these trips represent 4.5% of VMT.** The difference between the mode shift potential of car trips versus VMT occurs because short trips are more likely to have a competitive, non-car option.

This analysis shows that given existing travel patterns, even if everyone who could competitively switch to walking, biking and transit did switch, it would only achieve a quarter of the targeted 20% per capita VMT reduction. Therefore, policy-makers may wish to consider transportation infrastructure and service changes to further encourage mode shifts. To provide a starting point for those discussions, we evaluated the potential of five very broad scenarios to increase the competitiveness of alternative modes, and a sixth scenario that combines all of the previous five. The results show that **if we made it safe and comfortable to walk and bike everywhere it is legal, made e-bikes available, doubled transit service, and reduced car speeds, half of car trips and 16.5% of VMT could competitively shift to another mode.**

While the potential mode shifts in these scenarios are promising, it is important to remember that just because people *can* switch to another mode does not necessarily mean they *will* switch. The results also show that under existing conditions, people making 25% of car trips can switch to another mode but have not. More work is needed to assess how to most effectively prompt a shift to another mode once that mode is competitive. Pricing incentives or behavioral “nudges” are two strategies that might help.

Pricing incentives can include either strategies that make alternative modes less expensive—such as fare-free transit or vouchers to purchase an e-bike—or those that make driving more expensive—such as higher parking costs, gas prices, or tolls. If designed properly, pricing strategies have the potential to incentivize mode shift, reduce traffic congestion *and* generate revenue that can be used to make alternative modes better.

The concept of nudges comes from the field of behavioral economics and was popularized in a 2008 book by Thaler and Sunstein (Thaler and Sunstein 2008). The idea is to make small changes to the environment to make it more likely that people will make choices that benefit themselves or society. For example, a school cafeteria hoping to improve the dietary choices of students might put healthy food choices at eye level, while putting unhealthy foods in a less conspicuous location. In the transportation realm, nudges have been tested in research experiments that give travelers small gifts, tap into altruistic motives, or employ social pressure to incentivize them to make more sustainable travel choices, with modest effects (Hauslbauer et al. 2022; Steffen, Hook, and Witlox 2024; Riggs 2017).

Our analysis does not account for the changes to travel patterns that might be prompted by the proposed scenarios. If someone can safely bike to more places, they may choose more destinations within biking distance, rather than driving further to a location only accessible by car. Similarly,

someone who shifts from driving to transit for their work commute, may plan their shopping trips around that commute. The Metropolitan Council's activity-based travel demand model can simulate these travel pattern changes and estimate how many car trips and how much VMT are likely to switch to walk, bike or transit for each of the scenarios described here. Those model-based estimates would have trade-offs—they would have less network detail than this study and would be based in simulated rather than observed data. Thus, the data-driven and model-based estimates can be viewed as complements, each providing different evidence on the possible effectiveness of the proposed strategies.

Overall, we can draw several conclusions from this analysis:

- Biking has the highest potential for mode shift from car, and this potential can be greatly enhanced by building safe and comfortable bike infrastructure on all streets. However, only a small share of current trips that *could* be made by bike *are* made by bike, so that potential remains largely untapped. More work is needed to assess the extent to which this potential can be realized.
- Short trips have much higher mode shift potential. Compact land-use can provide a double-benefit of higher mode shift potential and shorter trips for those that remain in cars and is necessary to achieve greater VMT reduction. While this study is based on current land-use, policy-makers have some ability to guide how and where the Twin Cities region continues to grow in the coming decades.
- Even with broad changes to transportation infrastructure and service, many trips remain that cannot competitively shift from car. Vehicle electrification will be essential to reducing the greenhouse gas emissions from these trips, and policies aimed at incentivizing electric vehicles can be targeted towards travelers least able to switch modes and least able to afford an electric vehicle themselves. There are two ways in which electrification is a natural complement to mode shift. First, electric vehicle ownership rates are higher in suburban areas where the mode shift potential is less (Erhardt et al. 2024), partly because people living in single-family homes can more easily install in-home chargers than those living in apartments. Second, the mode shift potential is greatest in more urban areas, where people are more likely to live in apartment buildings and opportunities for off-street charging may be more limited. In these areas, planners can make informed decisions about how to allocate right-of-way and curb space across modes, such as choosing among car lanes, bike lanes and on-street chargers.

Mode shift, electrification, and compact land-use offer different levers for reducing the greenhouse gas emissions from car travel. The maps in Figure 13 show the locations with the highest mode shift potential, and conversely the locations where electrification may be necessary. These maps also show where future development can be concentrated to maximize the mode shift potential. We hope this information is valuable to planners and policy makers in the Twin Cities region as they consider how to respond to the climate crisis.

Acknowledgments and Code Availability

This study was performed by a team from the University of Kentucky (UK), the University of North Carolina at Chapel Hill (UNC), Metropolitan Council and its regional partners. Greg Erhardt (UK) was the study's principal investigator and the lead author of this report. Matt Bhagat-Conway (UNC) contributed to the study design, analysis and interpretation, and led the development of the routing methods. Ian Zhang (UK) led the software development for the analysis tool, and Rich Donohue (UK) contributed to visualization. Metropolitan Council staff involved included Johathan Ehrlich who was the project manager, Ashley Asmus who conceived of and initiated the study, and Liz Roten and Brandon Whited who contributed technically. Eric Lind from Metro Transit and Jim Henricksen from the Minnesota Department of Transportation provided review and input.

This study was funded by Metropolitan Council through contract number 22P159 with the University of Kentucky Research Foundation.

The software tools used to conduct the analysis, and an interactive data summary tool are available at: <https://github.com/Metropolitan-Council/vmt-mode-shift-study>. This code repository is planned for public availability following a privacy review. The software tools are suitable for application to additional years of the TBI, or similar data in other regions. Documentation on using the software tools is available in the repository, starting with the README.md file.

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Appendix A: TBI Data Processing

The VMT Reduction Mode Shift Study relies on an analysis of real-world trips reported in the 2019 and 2021 Travel Behavior Inventory (TBI) surveys of the Minneapolis/St. Paul Region. This technical memorandum describes how those trips were processed to support this study.

That processing is completed in the accompanying notebook **Clean and Code TBI.ipynb** stored in the project github repository. The result of that processing is the file **tbi_merged.csv** which is stored on the project Teams site. The resulting data file contains one record for each linked trip to be analyzed for its mode shift potential, along with associated household, person and day attributes.

Main Data Processing

For this study, we include data from both the 2019 (wave 1) TBI and the 2021 (wave 2) TBI. We expect future waves to be added as they become available, allowing for changes to be tracked over time.

Each TBI is a week-long, smartphone-enabled household travel survey that combines passive tracing of participants smartphone locations with active responses to questions, particularly on mode and purpose. Please refer to the survey documentation for details on its implementation and initial processing. Each wave of TBI data is provided in several main files:

- hh.csv: One record for each household surveyed.
- person.csv: One record for each person in those households.
- day.csv: One record for each travel day, generally 7 days per person.
- trip.csv: One record for each unlinked trip made by those travelers.
- location.csv: A series of point locations and timestamps for each trip from the smartphone traces. These files are not processed further, and will only be used in this study to QA/QC the routes.
- vehicle.csv: One record for each vehicle in the household. This will us to link VMT reductions to the emissions reductions associated with specific vehicles. Ignore vehicle_id in the Wave 2 trip list and instead use mode_type_detailed for matching.

The core data processing in this task involves:

1. Merging each of the the household, person, day and trip files across the survey waves and rectifying any differences in the available fields.
2. Retaining only the fields potentially relevant to this study.
3. “Decoding” the data by converting numerical codes into strings with a more obvious interpretation. For example, 995 might be replaced by “Missing” such that 995 is not inadvertently included in an average.
4. Aggregating from unlinked trips to linked trips.
5. Determining the mode and reference VMT.
6. Excluding records that are out of scope or missing important data.

The details are explicit in the data processing notebook. Several of the steps warrant additional explanation here, as well as the question of how to use the weights in application.

Trip Linking

When a person makes a trip by transit, they usually combine multiple modes, such as walking and riding the bus, and often have a stop to change modes while they wait for the bus. These “change mode” stops are coded as trip ends in the TBI data. However, if the person were to use a different mode, they would not need to make those stops and could proceed directly to their destination. For that reason, the TBI data identifies a “linked trip” ID allowing us to identify how those individual trips combine for an overall origin to destination.

A similar problem arises for car trips when someone stops for gas. The stop for gas is usually on the way to another destination, and would not be necessary if the trip were not made by car. Therefore, we added logic to link out stops for gas when they are part of a longer chain of trips. We keep gas stops when they are the only non-home activity on a tour (home - gas - home).

In addition to stops for gas, the data show that a small number of trips on modes other than transit are also linked. Spot checking showed that these appear to be reasonable, such as someone reporting changing modes from what appears to be a church van to a personal vehicle.

Once the linked trips are identified, we merge them to create one record for each linked trip. The origin and departure time are from the first trip. The destination and arrival time are from the last trip. We sum the total trip duration and distance, then re-calculate the speed. The linked trip’s mode is calculated based on a priority system, which from highest to lowest priority is:

1. Long distance passenger mode
2. School bus
3. Transit
4. Car
5. Taxi/Ridehail/Carshare
6. Bike/Scooter
7. Walk
8. Missing/Other

For example, if a linked trip involves car-transit-walk, the linked trip is coded as a transit trip because transit is the highest priority mode.

Mode and Reference VMT

The available modes coded in the survey are somewhat more complicated, but we code them into the 7 modes shown above for this study. We do so using the following equivalency:

Detailed Mode	Mode
Long distance passenger mode	Long distance passenger mode
School bus	School bus
Ferry	Transit
Rail	Transit
Public bus	Transit
Other bus	Transit
Shuttle	Transit
Transit	Transit
Household vehicle	Car
Other vehicle	Car
Vehicle	Car
Taxi	Taxi/Ridehail/Carshare
Carshare	Taxi/Ridehail/Carshare
For-hire vehicle	Taxi/Ridehail/Carshare
Smartphone ridehailing service	Taxi/Ridehail/Carshare
Smartphone-app ride-hailing service	Taxi/Ridehail/Carshare
Bicycle or e-bicycle	Bike/Scooter
Bike-share	Bike/Scooter
Scooter-share	Bike/Scooter
Micromobility	Bike/Scooter
Walk	Walk
Missing: Non-response	Missing/Other
Missing	Missing/Other
Other	Missing/Other

To calculate the total VMT in the region, and therefore the potential VMT reduction, we need to be cognizant of the vehicle occupancy. Each record in our resulting data set is a person trip, but it is not necessarily a vehicle trip. We calculate a vehicle trip field as follows: if the mode is Car or Taxi/Ridehail/Carshare, the number of vehicle trips is $1 / \text{num_travelers}$. If the number of travelers is missing or unknown, we assume it is 1. If the mode is not Car or Taxi/Ridehail/Carshare, then it becomes 0 vehicle trips. The VMT of the record is calculated as the vehicle trips times the trip distance.

With this accounting, if three people travel together 1 mile by car, they produce 1 vehicle trip, and 1 VMT. If all three bike instead, we would reduce VMT by 1. If only one bikes, our analysis would show a VMT reduction of 1/3 of a mile. While this is not entirely realistic, all three would have the same alternative routes, so would shift together in our analysis unless there is a constraint that applies to some, but not others.

Exclusions

The scope of this analysis is trips made on a typical 7-day week by people in households in the 19-county study area, where both ends of the trip are within that study area. We separately report these data for a 2019 and 2021 baseline. To align with this scope, we exclude records as follows:

Exclusion	Trips Excluded	Percent of VMT Excluded
Latitude, longitude or timestamp is missing.	1,656	0%
Home location outside study area.	0	0%
Origin or destination outside study area.	37,520	21.5%
Trip on a long-distance passenger mode.	166	0%
Mode is missing.	15,284	0%
Distance is less than 0.01 miles (52.8 ft).	374	0%
Distance is more than 100 miles (5280 ft).	243	3.3%
Detailed mode is ferry	12	0%

The exclusions are applied in the order shown—the numbers reported do not consider that more than one exclusion may apply. The VMT is calculated as described above, which explains why the VMT reduction from excluding missing mode trips is zero.

It is notable that almost 25% of the VMT in the survey has either one end outside the region or is trips longer than 100 miles. Assuming the trip distances are coded correctly (it would be worth checking a few if we were to do something other than exclude these trips), it highlights how important long-distance travel may be to VMT reduction efforts.

In addition, the data do not include trucks or commercial vehicles, which also generate substantial VMT. We think it is appropriate to exclude them from this study, but it is worth keeping in mind that there is a substantial portion of VMT that we will not shift.

Weighting

The data include four weights: household, person, day and trip. Please refer to the TBI documentation for details on how they are calculated. Each is an expansion factor, which means that when it is applied the data are “expanded” to represent the full population of the study area. For the household and person weights, this aligns with the number of households and the population of the study area. The day and trip weights are expanded to represent average weekday (Monday through Thursday) conditions.

For this study, we recommend using the day weight. When applied, the data should expand to roughly the VMT over the course of a typical weekday, accounting for people who reported different numbers of days of travel. In the future, we might consider weekend VMT, but this requires some more thought about how to adjust the weights.

Appendix B: Routing Methods

The goal of the re-routing analysis is to determine the route details for each trip if it had been taken by a different mode—for instance, how long a driving trip would have taken by cycling, or how many transfers would be required if a trip were shifted to transit from another mode. We define a “profile” for each alternate mode modeled, as well as for each scenario. These profiles contain the detailed parameters needed to find the paths we have determined to be best. For instance, the walking profile defines how quickly people walk, how far out of their way they will go to travel on a flatter or safer route, and so on.

These profiles are used to determine the weights which will be used in routing. Each street segment is assigned a weight, based on its length, steepness, safety, and anything else the profile deems relevant to route choice. The route that is found is the one where the weights of all constituent street segments sum to the lowest amount.

We have four baseline profiles: walking, cycling, driving, and transit. We additionally have profiles for the scenarios described above: walking with no safety constraints, biking with no safety constraints, e-biking, and lowered car speeds. The doubled transit frequency scenario is not a separate profile, but simply modified GTFS input data. Street routing uses [OSRM](#) and the [OSRM.jl Julia package](#). Transit routing uses the [TransitRouter.jl Julia package](#). In this section, we summarize the profiles and the justification for choosing the software we chose; more detailed technical information, instructions for running the routing, and minor errata are available in the routing documentation within the repository itself.

There were a number of reasons why we chose OSRM, OSRM.jl, and TransitRouter.jl for this project. The first consideration in choosing a routing package was customizability; we wanted to be able to customize the routing to use congestion data from StreetLight, and to account for stress and elevation in bike/walk routing.

The second consideration was whether to adopt a *single-objective* or *multi-objective* street routing engine. A single-objective routing engine finds routes that minimize a single quantity. This could be time, distance, or a *generalized cost* which consists of a weighted sum of multiple components such as time, slope, safety, etc. A multi-objective algorithm, in contrast, will find multiple routes that represent possible tradeoffs between different objectives. For instance, it might find a shorter route that traverses a steep hill, a much longer but relatively flat route, and a compromise between the two.

While a multi-objective routing algorithm is theoretically attractive, we chose to use a single-objective algorithm for two reasons. Firstly, most existing off-the-shelf routing software uses single-objective routing algorithms. Secondly, multi-objective routing is significantly more computationally complex and therefore slower than single-objective routing, a challenge given the size of the TBI dataset.

We considered four software options for street routing and two for transit routing. For street routing, we considered OSRM, Valhalla, pandana, and r5r. The performance of pandana was determined to be insufficient. Customizability for Valhalla and r5r was limited and required programming in C++ or Java (respectively), which was undesirable for allowing Metropolitan Council to continue development of the project after the conclusion of the initial grant. OSRM has very high performance, and routing weights are customizable using Lua, a fairly straightforward scripting language. One of the project team members, Matt Bhagat-Conway, already had experience with OSRM, as well as a Julia wrapper

to allow computationally efficient access from a dynamic programming language, further tipping the scales towards OSRM.

For transit routing, we considered `r5r` and `TransitRouter.jl`. We ultimately chose `TransitRouter.jl`, because it provides more details about each individual trip than `r5r`; `r5r` is more intended for accessibility analysis than routing of individual trips. While `r5r` does have routing for individual trips, it uses a somewhat slower algorithm that uses heuristics to find both optimal and slightly sub-optimal trips. Furthermore, `TransitRouter.jl` integrates directly with OSRM, allowing walking to and from transit to use the same routing profiles as walk-only trips.

Having chosen software, we needed to define profiles to use in routing. The walk profile is based primarily on travel time and safety. Hardy, Bell, and McCahill (2019) develop a set of equivalent speeds to adjust the value of pedestrian infrastructure based on quality, shown in the table below **Error! Reference source not found.** We use these penalties to adjust the routing *weights* but not the speeds, as people don't actually walk slower on poorer facilities (likely the opposite). We find the minimum weight path based on these weights, but still calculate speed without regard to roadway quality. Vehicle speeds are based on OSM maxspeed and OSRM defaults, rather than observed speeds from StreetLight, to match the methodology of Hardy, Bell, and McCahill (2019) and due to OSRM technical limitations. For ways that do not have specified max speeds in OpenStreetMap, we use the OSRM default speeds for these ways.

Weight penalties by roadway type					
Functional class	Lanes	Has sidewalk	Vehicle speed	Pedestrian speed	Weight penalty
Any	Any	Yes	> 55 mph	1.5 mph	2
Any	6+	Yes	≤ 55 mph	2.4 mph	1.25
Freeway/major highway	4–5	Yes	≤ 55 mph	2.4 mph	1.25
Major arterial	4–5	Yes	≤ 55 mph	2.4 mph	1.25
Minor arterial/collector	4–5	Yes	41-55 mph	2.4 mph	1.25
Minor arterial/collector	4–5	Yes	≤ 40 mph	2.7 mph	1.11
Local	4–5	Yes	41-55 mph	2.4 mph	1.25
Local	4–5	Yes	31-40 mph	2.7 mph	1.11
Local	4–5	Yes	≤ 30 mph	3.0 mph	1
Freeway/major highway	≤ 3	Yes	≤ 55 mph	2.4 mph	1.25
Major arterial	≤ 3	Yes	31-55 mph	2.4 mph	1.25
Major arterial	≤ 3	Yes	≤ 30 mph	2.7 mph	1.11
Minor arterial/collector	≤ 3	Yes	41-55 mph	2.4 mph	1.25
Minor arterial/collector	≤ 3	Yes	≤ 40 mph	2.7 mph	1.11
Local	≤ 3	Yes	41-55 mph	2.4 mph	1.25
Local	≤ 3	Yes	31-40 mph	2.7 mph	1.11
Local	≤ 3	Yes	≤ 30 mph	3.0 mph	1
Any	4+	No/Unknown	Any	1.5 mph	2
Any	≤ 3	No/Unknown	> 30 mph	1.5 mph	2
Freeway/major highway	≤ 3	No/Unknown	≤ 30 mph	1.5 mph	2
Major arterial	≤ 3	No/Unknown	≤ 30 mph	2.7 mph	1.11
Minor arterial/collector	≤ 3	No/Unknown	≤ 30 mph	2.7 mph	1.11
Local	≤ 3	No/Unknown	≤ 30 mph	2.7 mph	1.11
Off-street	N/A	N/A	N/A	3.0 mph	1

Walking on mild hills is not that bad, but very steep hills are unpleasant. We apply a double weight to streets or portions of streets sloped more than 10%.

In the walk scenario, we remove all penalties for walking on less comfortable streets, and assume it is safe and comfortable to walk anywhere it is legal to do so.

The bicycle profile uses an estimated Level of Traffic Stress derived from OpenStreetMap. We calculate LTS using a methodology based on one used by the [Accessibility Observatory](#). We only allow biking on LTS 1 and 2 streets, with a 10% speed and weight penalty for LTS 2. On LTS 3 and 4 streets, we require walking bikes, using the speeds derived from the walking profile. Biking has a more significant penalty for slopes. We treat each meter of elevation gain as equivalent to 25 flat meters when biking (or when walking a bike).

The bike scenario reassigns all streets to LTS 1. The e-bike scenario additionally removes elevation penalties, and increases speeds 13%.

Auto routing uses speeds based on StreetLight data for the appropriate time period and day of week (weekdays, Saturdays, and Sundays), if available. If StreetLight data are not available for a particular segment, we fall back to the OSRM default of 80% of the tagged max speed, or a default speed based on roadway type (see table below).

Default OSRM speeds		
OSM type	Speed (km/h)	Speed (mph)
motorway	90	56
motorway_link	45	28
trunk	85	53
trunk_link	40	25
primary	65	40
primary_link	30	19
secondary	55	34
secondary_link	25	16
tertiary	40	25
tertiary_link	20	12
unclassified	25	16
residential	25	16
living_street	10	6
service	15	9

The car scenario slows cars down based on NACTO guidance for urban streets. The NACTO guidance does not have guidelines for freeways, so we set guidelines based on local context. Speed limits used are presented below. We assume average travel speeds are 80% of the NACTO recommended speed limits, consistent with how OSRM handles posted speed limits in the base case.

	NACTO speed limits		
	Urban	Suburban	Rural
Shared street/alley	10 mph	10	10
Minor streets	20	20	20
Major streets	25	35	35
Freeways	55	55	65

Transit routing is based on fall 2019 (pre-pandemic) GTFS for Metro Transit and MVTA. For access to transit, we use our baseline walk network. Transit is more time-dependent than the other modes; exactly when you leave may result in a very different trip. For most trips, we route “forward” - that is, we constrain the departure time. For some trips, we route in “reverse,” constraining the destination of the trip. This is based on the trip purpose priority, listed below:

1. Medical visit (e.g., doctor, dentist)
2. Errand with appointment (e.g., haircut, accountant)
3. Errand with appointment (e.g., haircut)
4. K-12 school
5. Attend K-12 school
6. Daycare or preschool
7. Attend daycare or preschool
8. College/university
9. Attend college/university
10. Other education-related (e.g., field trip)

11. Attend other education-related activity (e.g., field trip)
12. Attend vocational education class
13. Other type of class (e.g., cooking class)
14. Attend other type of class (e.g., cooking class)
15. Primary workplace
16. Work-related activity (e.g., meeting, delivery, worksite)
17. Went to work-related activity (e.g., meeting, delivery, worksite)
18. Other work-related
19. *All others*

Whichever end has the higher priority is the end that is constrained - so a trip from home to work would be constrained at work, while a trip from a medical visit to work is constrained at the medical visit. We assume that even constrained trip purposes have some flexibility. We allow violating the departure or arrival time constraint for all trip purposes by up to five minutes if it makes the trip at least 15 minutes better. We use the range-RAPTOR algorithm to find latest-departure-given-earliest arrival trips (Delling, Pajor, and Werneck 2015; Bhagat-Conway 2023). There are often multiple trips that arrive at the same time due to transferring to the same vehicle; the latest-departure-given-earliest arrival trip is the one of these that leaves the latest.

During routing, we apply a walk limit of 1.5 miles for access, egress, and transfers. In the analysis, results were further filtered to a shorter walking distance. Because of this, some trips may be filtered out at analysis time if they require walking more than 0.9 miles, even if there was a slower option that required less walking. This has been deemed an acceptable tradeoff to allow changing the walk cutoff without re-running transit routing.

The transit scenario doubles frequency on all routes. The GTFS files are coded with actual timetables, not frequency entries, so we create new GTFS files with new trips. We do this by duplicating each trip and inserting it halfway between the original trip and the next trip on the route. This means that changes in run time over the course of the day will be reflected in the output.

Appendix C: Terminal Times

Terminal time is intended to count the amount of time it takes a person to park their car, and walk from their car to their actual destination. If someone is parked in their driveway, it might take a minute or less to walk to the car, unlock it, start it, and pull out of their driveway. If instead, someone parks in a large parking lot at a shopping mall, it might take a few minutes to park and walk to the mall's entrance. Walking and transit trips do not incur terminal time, but bike trips do because cyclists must park and lock their bike. Terminal times can have an important effect on the competitiveness of driving versus other modes, particularly in downtown areas where parking is more scarce and for short trips where the terminal time might constitute a larger share of the trips travel time.

Despite its importance, terminal time is rarely recorded in standard data sets. We examined whether it might be possible to derive the terminal time from the TBI survey data, but could not do so reliably. Finding a way to measure terminal time would be a valuable topic for future research. In this study, we instead make the following assumptions, which are specific to the traffic analysis zones (TAZs) for each trip's origin and destination.

Category 1: 5 minutes of terminal time. TAZs where Population + Employment per Acre > 80 . This threshold includes:

- Downtown Minneapolis
- Downtown St. Paul
- University of Minnesota Campus
- Mall of America
- MSP Airport (added manually to avoid low density of runways)
- Abbott Northwestern Hospital

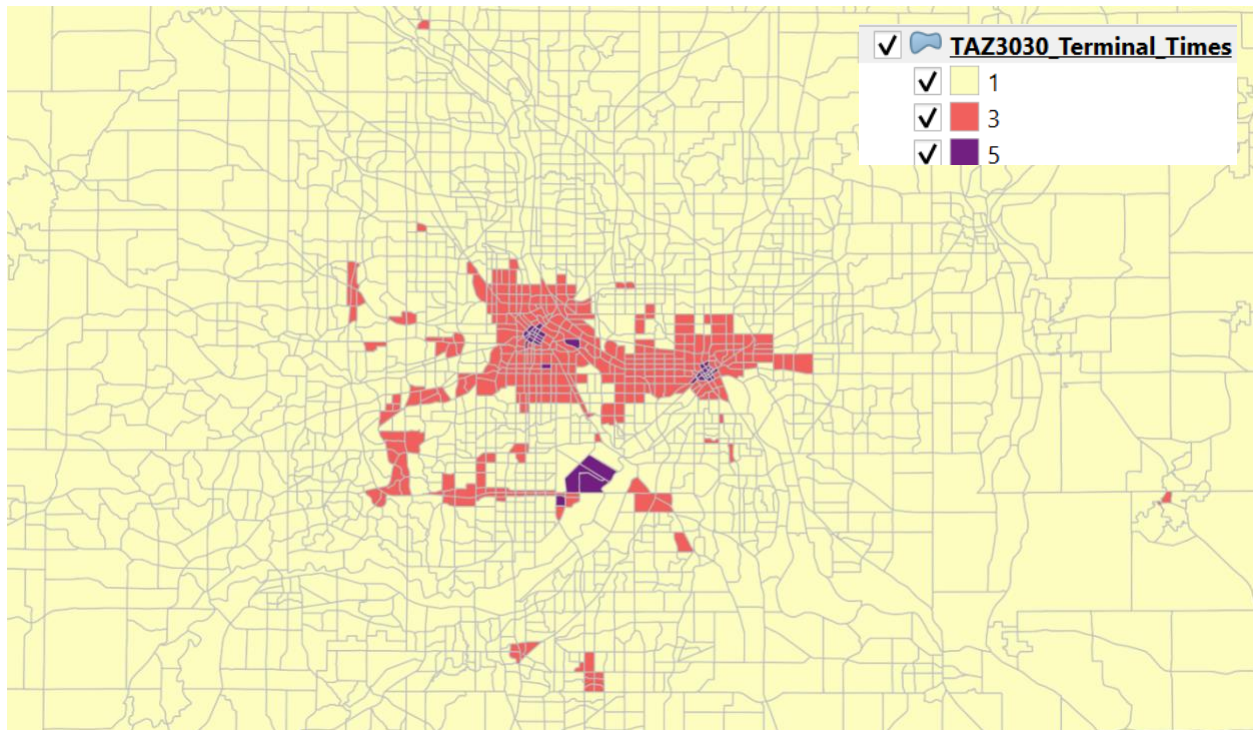
Category 2: 3 minutes of terminal time. TAZs where Population + Employment per Acre > 30 and ≤ 80 . These thresholds cover:

- Most of Transit Service Area 1, as defined by Metropolitan Council, which are denser urban neighborhoods.
- It also includes suburban commercial areas, such as shopping malls, WalMarts, and a few major employers, such as Thompson Reuters. Broadly, these are areas with very large parking lots, where it may take some time just to find a parking spot and walk across the lot.

Category 3: 1 minute of terminal time. All other areas.

After the initial screening, we undertook a manual smoothing process to ensure the results looked sensible. We focused on cases where one TAZ was classified differently from neighboring TAZs, and considered whether the result was logical, and aligned with what we observed in Google maps imagery. The map below shows the resulting terminal times by TAZ.

For bike trips, we assume 1 minute of terminal time at the origin and 1 minute at the destination.



Car terminal times assumed for the Twin Cities region

Appendix D: Feasibility Analysis

We consider which car trips could *feasibly* switch to another mode, given the possible routes. For example, it might be *possible* to walk 10 miles to work, but it is not necessarily *feasible* to do so. We determine the feasibility of shifting to another mode by introducing the “5% rule”:

If we have a reason to think that something is a barrier to shifting modes AND fewer than 5% of observed trips currently using that mode violate that barrier, then we consider doing so to be infeasible.

We also consider whether a trip can be completed within the available time without interfering with a work, school or escort passenger activity. For example, an afternoon trip between finishing work and picking up a child from daycare likely has limited flexibility on both ends, and there may not be sufficient time available to accommodate a longer travel time.

The following pages report our findings of this feasibility analysis. We report both factors which we found to be constraints, and those which we considered but did not include as constraints because they did not meet the 5% rule.

Feasibility of shifting to walk

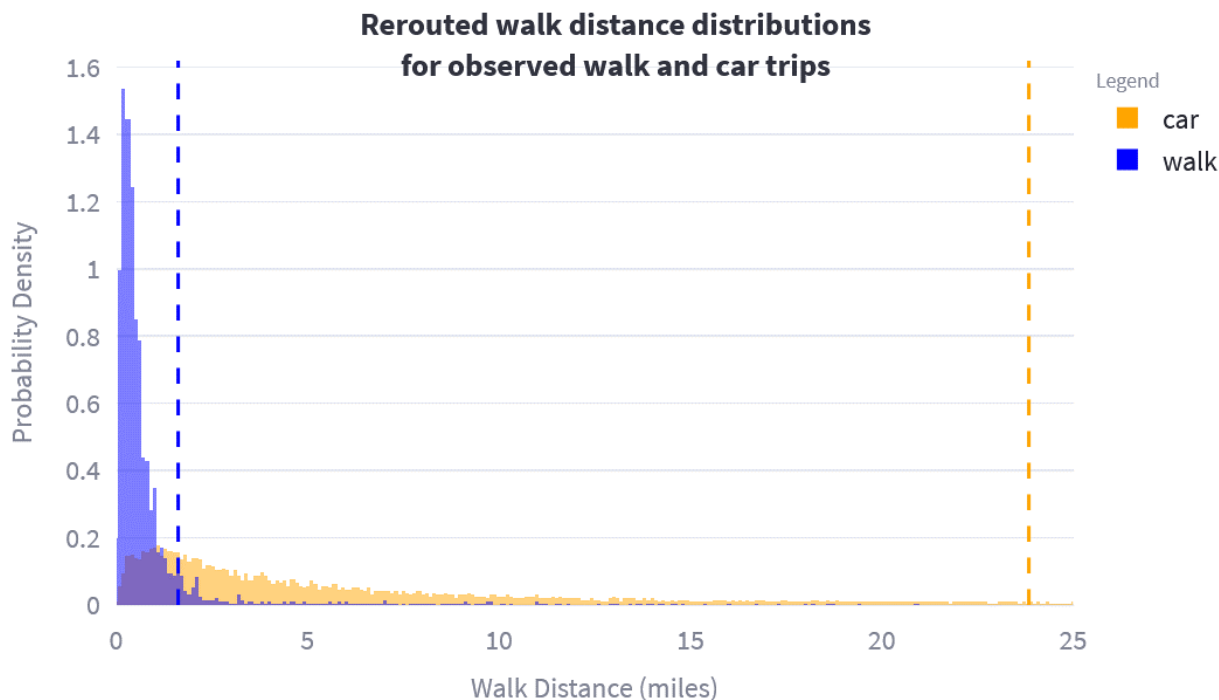
We identified two constraints limiting the feasibility of walking: walk distance and walk timing. We also considered the trip purpose and two measures of weather (precipitation and temperature) as possible constraints, but did not find them warranted.

Constraint: Feasible walking distance

Walk trips are generally short. Here we consider the trip length distribution of walking trips, and set a maximum allowable distance to be considered feasible to walk. **We use the 95th percentile of the distances of the observed walking trips, or 1.6 miles, and consider walking to be infeasible if the walking distance exceeds this value.**

In this figure, the distributions for walking distances from the rerouting analysis for observed walk trips can be seen, alongside the distributions for calculated re-routing walking distances for observed car trips. It is readily seen that the car trip walking distances are far more right-skewed than the walking trip walking distances, which is to be expected, as car trips are more likely to be used for long distance trips.

Of note are the two lines present in the graph, which represent the percentiles of the data specified as a parameter. In particular, any part of the orange distribution to the left of the blue line is within the threshold for feasibility (for the given percentile cutoff) and thus constitutes a feasible mode shift.

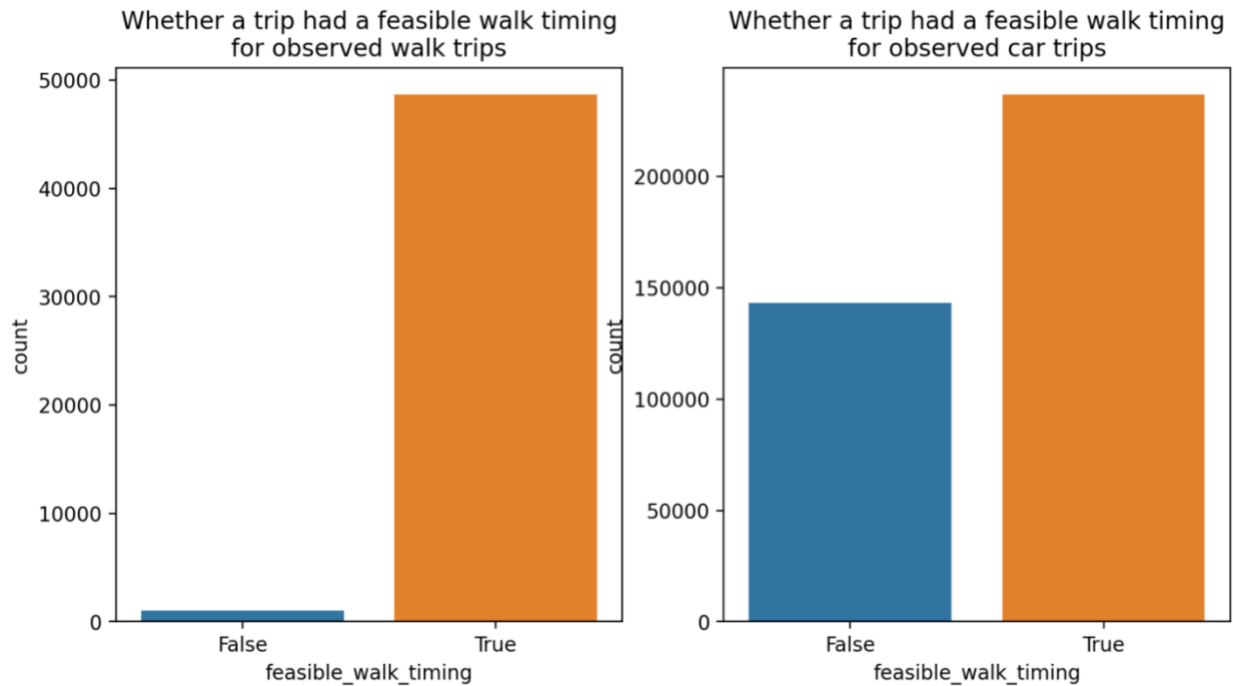


Before this step, **100.0%** of trips could shift to walk feasibly, and after this step, **22.8%** of trips could shift to walk feasibly. Additionally, before this step, **100.0%** of VMT could be mitigated with shifts to walk, and after this step, **3.0%** of VMT could be mitigated with shifts to walk. Overall, independent of other steps, **22.8%** of all car trips and **3.0%** of VMT can satisfy this constraint.

Constraint: Feasible walk timing

For some people, it is infeasible to switch from a car mode to an alternative mode due to timing constraints; with the longer durations of alternative modes, a person may not have the ability to do everything they need to in a given day. **Here we consider whether a trip can be completed by walking within the time available between visiting subsequent out-of-home activities.** With this definition, the first and last trip of the day always have feasible timing because a person can simply depart sooner or arrive later. To provide some flexibility, we allow the traveler to depart up to 5 minutes early and arrive up to 5 minutes late for an out-of-home activity. If a trip cannot be completed by walking within the available time, a traveler would have to reschedule other activities—something which is theoretically possible but beyond the scope of this study.

Here, a bar plot of the result of the timing logic when applied to walk shifts can be seen.



Before this step, **22.8%** of trips could shift to walk feasibly, and after this step, **18.0%** of trips could shift to walk feasibly. Additionally, before this step, **3.0%** of VMT could be mitigated with shifts to walk, and after this step, **2.2%** of VMT could be mitigated with shifts to walk. Overall, independent of other steps, **63.0%** of all car trips and **57.0%** of VMT can satisfy this constraint.

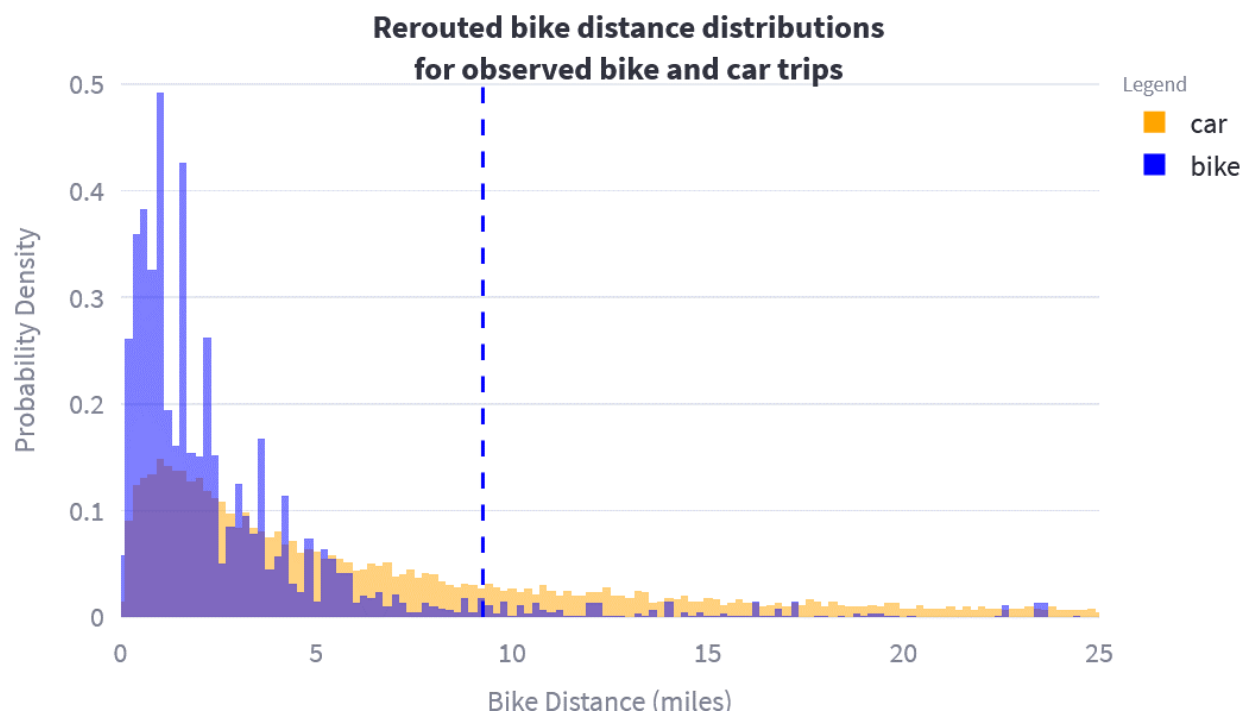
Feasibility of shifting to bike

We identified four constraints limiting the feasibility of biking: distance, snow depth, percent of distance on high level-of-traffic stress facilities, and timing. We also considered the trip purpose, precipitation and temperature as possible constraints, but did not find them warranted.

Constraint: Feasible biking distance

Bike trips are generally on the shorter side. Here we consider the trip length distribution of biking trips, and set a maximum allowable distance to be considered feasible to bike (as before, this can be specified in the parameter section). By default, the maximum allowable distance is the 95th percentile of distances of the observed biking trips. **We use the 95th percentile of the distances of the observed bike trips, or 9.3 miles, and consider biking to be infeasible if the biking distance exceeds this value.**

In this figure, we can see a comparison between the distributions of biking distance for observed biking trips versus biking distance if observed car trips were to shift to biking. The car distance distribution is more right-skewed, which is reasonable as cars generally go longer distances. The line here represents the specified percentile of each distribution, and all trips of the orange distribution to the left of the blue line can feasibly switch to biking, as their biking distance is reasonable compared to observed biking distances.



Before this step, **100.0%** of trips could shift to bike feasibly, and after this step, **67.3%** of trips could shift to bike feasibly. Additionally, before this step, **100.0%** of VMT could be mitigated with shifts to bike, and after this step, **26.9%** of VMT could be mitigated with shifts to bike. Overall, independent of other steps, **67.3%** of all car trips and **26.9%** of VMT can satisfy this constraint.

Constraint: Bike snow depth

Snowing can impact the feasibility for various modes significant, but this effect is particularly pronounced for biking. This is due to the snow decreasing the traction a bike can get to maneuver and

due to the discomfort that would arise from traveling through freezing temperatures at fast speeds, exposed fully to the elements.

Snow depth is measured in mm and is defined by NOAA (<https://www.weather.gov/gsp/snow>) as the total depth of snow, ice pellets, or ice on the ground at the time of observation, gauged using a measuring stick. This figure is meant to represent the average depth of snow/ice at ground level at the usual measurement site. (Snowfall is measured using a snowboard with respect to the previous observation--is accumulation of snow over a day).

Below, the summary statistics for the snow depth (in mm) that was present during the day of the trip when segmented by observed TBI mode is shown. It should be noted that at the default 95th percentile parameter, the snow depth for observed bike trips is 0 while this is not the case for observed car trips, indicating bike trips are more influenced by deeper snow than car trips are. **Therefore, we consider biking to be infeasible when there is snow on the ground.**

	Car: snow depth	Walk: snow depth	Bike: snow depth	Transit: snow depth
Count	380027.000000	49670.000000	7692.000000	11269.000000
Mean	4.594439	2.687739	0.536531	4.547431
Std	11.011706	8.160208	3.578943	10.491438
Min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000
75%	3.000000	0.000000	0.000000	3.000000
95%	36.000000	20.000000	0.000000	33.000000
Max	53.000000	53.000000	53.000000	53.000000

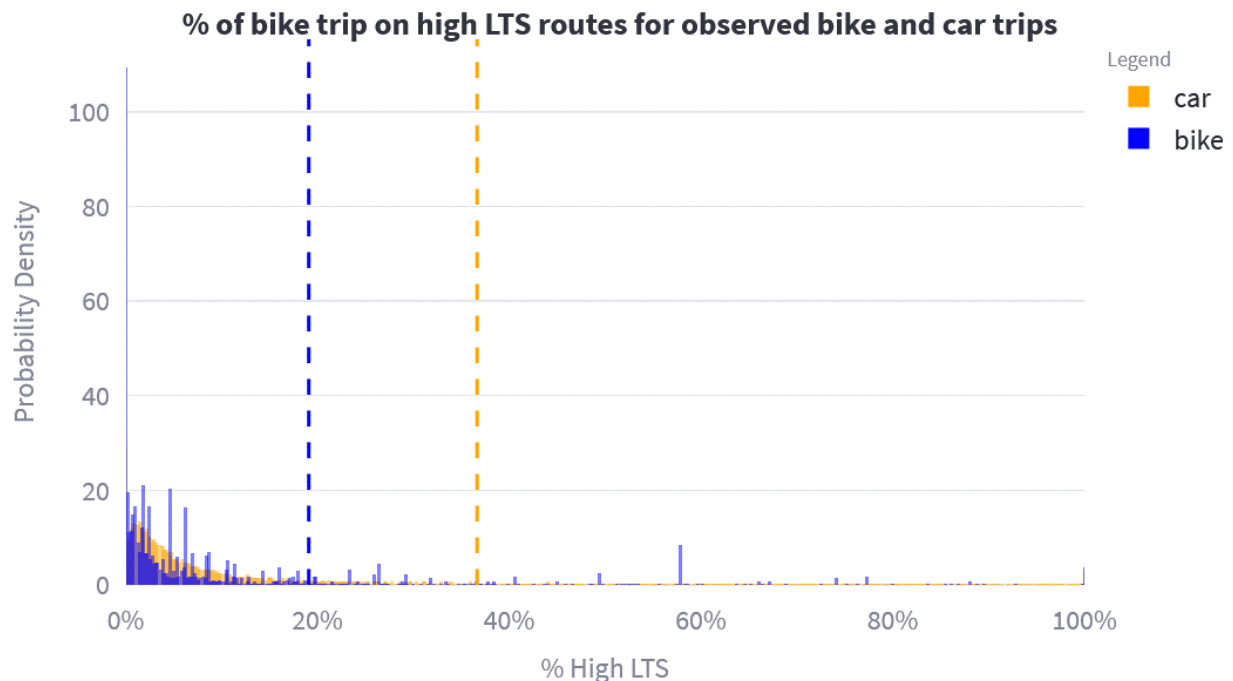
Before this step, **67.3%** of trips could shift to bike feasibly, and after this step, **51.5%** of trips could shift to bike feasibly. Additionally, before this step, **26.9%** of VMT could be mitigated with shifts to bike, and after this step, **20.6%** of VMT could be mitigated with shifts to bike. Overall, independent of other steps, **76.6%** of all car trips and **76.4%** of VMT can satisfy this constraint.

Constraint: Percent of biking distance on high LTS facilities

The level of traffic stress (LTS) quantifies how stressful/difficult it is to bike in a location, ranging from places with dedicated bike lanes at LTS 1 to main streets without any developed biking infrastructure at LTS 4. LTS 1 streets are considered safe and comfortable by almost all riders, LTS 2 for most adults, whereas LTS 3 and 4 are more stressful.

The distance through high traffic stress locations an individual would have to bike through to reach a destination is thus likely strong indicator for the feasibility that a trip could switch to biking. Here, we consider the percent distance traveled during a trip on the higher LTS categories, 3 and 4. We consider the percentage in particular because car trips are naturally longer than biking trips and distances are already accounted for, so percentages even things out for consideration.

It is very clear here from this figure that observed bike trips tend to be far more left-heavy for high stress distance than car trips (if they were to switch to biking), which makes sense as the reason these bike trips are observed is because they are feasible. The lines represent the specified percentiles for each of the two distributions, and any part of the orange distribution to the left of the blue line can feasibly switch to biking, under the high stress biking distance feasibility indicator.

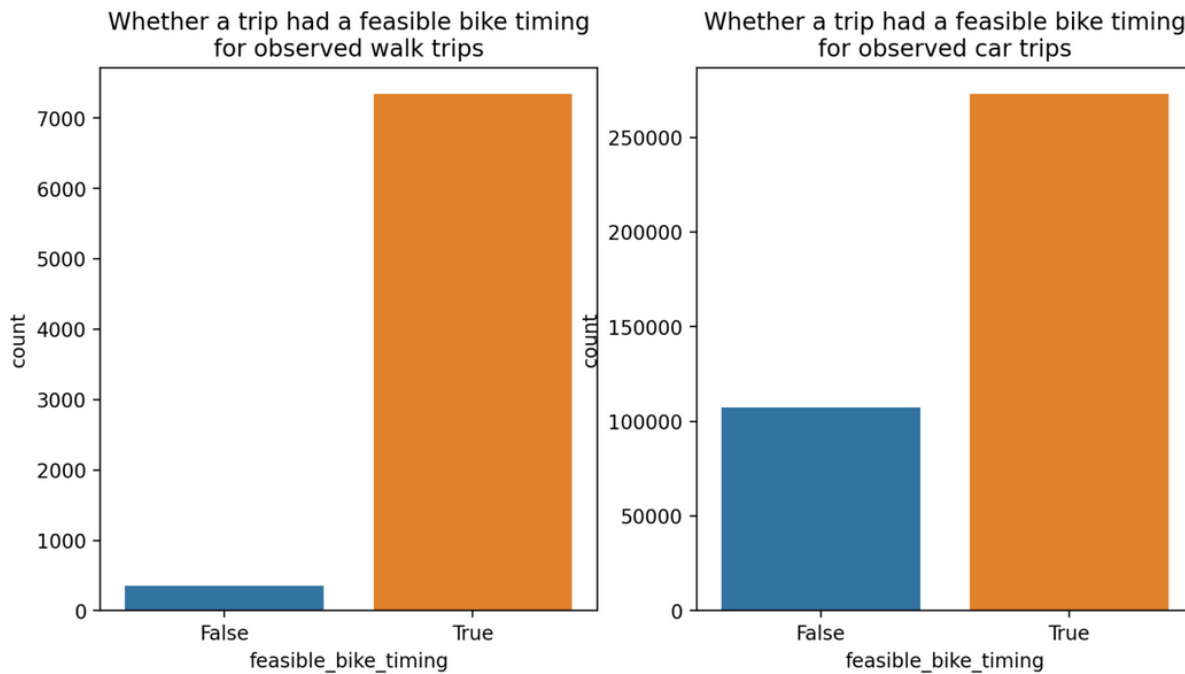


The results show that only 5% of observed bike trips have more than 20% of their distance traveled on LTS 3 or 4 facilities, so we consider bike routes that exceed this share to be **infeasible**. Before this step, 51.5% of trips could shift to bike feasibly, and after this step, 43.0% of trips could shift to bike feasibly. Additionally, before this step, 20.6% of VMT could be mitigated with shifts to bike, and after this step, 18.6% of VMT could be mitigated with shifts to bike. Overall, independent of other steps, 86.7% of all car trips and 93.9% of VMT can satisfy this constraint.

Constraint: Feasible bike timing

For some people, it is infeasible to switch from a car mode to an alternative mode due to timing constraints; with the longer durations of alternative modes, a person may not have the ability to do everything they need to in a given day. **Here we consider whether a trip can be completed by biking within the time available between visiting subsequent out-of-home activities.** With this definition, the first and last trip of the day always have feasible timing because a person can simply depart sooner or arrive later. To provide some flexibility, we allow the traveler to depart up to 5 minutes early and arrive up to 5 minutes late for an out-of-home activity. If a trip cannot be completed by biking within the available time, a traveler would have to reschedule other activities—something which is theoretically possible but beyond the scope of this study.

Here, a bar plot of the result of the timing logic when applied to bike shifts can be seen.



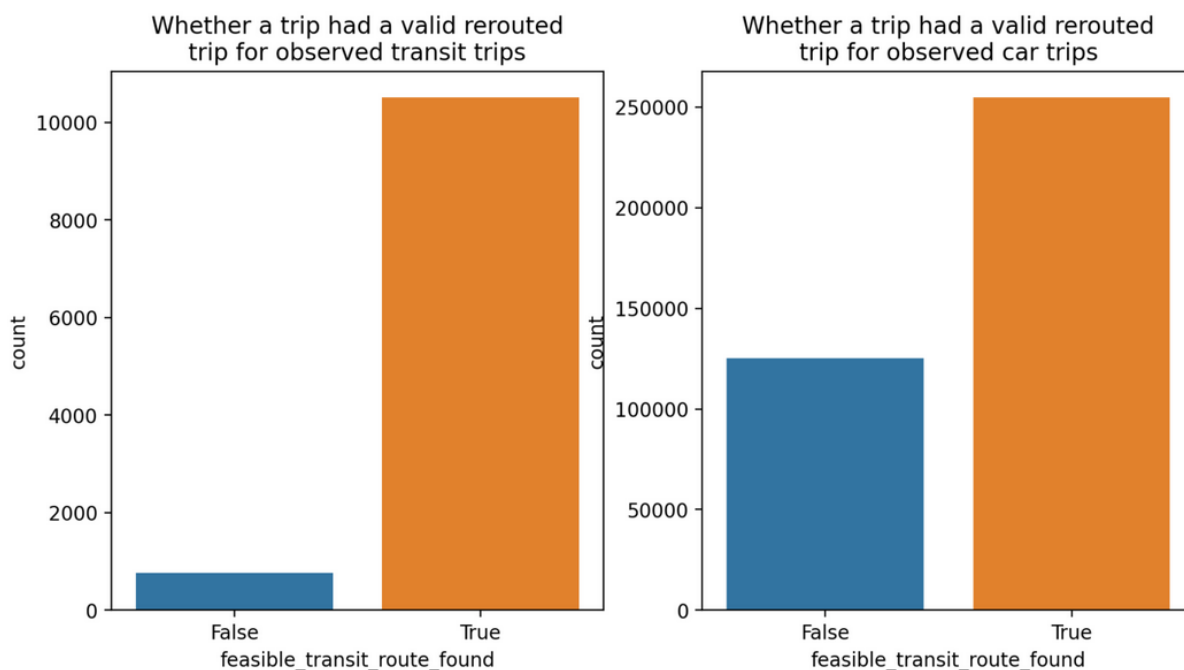
Before this step, **43.0%** of trips could shift to bike feasibly, and after this step, **32.5%** of trips could shift to bike feasibly. Additionally, before this step, **18.6%** of VMT could be mitigated with shifts to bike, and after this step, **13.0%** of VMT could be mitigated with shifts to bike. Overall, independent of other steps, **71.4%** of all car trips and **63.7%** of VMT can satisfy this constraint.

Feasibility of shifting to transit

We identified several constraints limiting the feasibility of shifting to transit: number of transfers, access distance, egress distance, waiting time and timing. We also considered the trip purpose as a possible constraint, but did not find it warranted.

Constraint: Valid transit path found with 2 or fewer transfers

The TBI data record the number of transfers for transit trips, and the data show that fewer than 5% of observed transit trips involve more than 2 transfers—therefore, we consider more than 2 transfers to be infeasible for new tips. We impose the transfer limit (and a loose limit on walking distance) in the routing method itself to avoid creating a large number of complex and infeasible paths, so the subsequent constraints rely on first meeting this constraint. Because we build this constraint into the routing itself, our analysis simply records whether a valid route path was found. The charts below summarize these results.



Before this step, **100.0%** of trips could shift to transit feasibly, and after this step, **61.4%** of trips could shift to transit feasibly. Additionally, before this step, **100.0%** of VMT could be mitigated with shifts to transit, and after this step, **47.5%** of VMT could be mitigated with shifts to transit. Overall, independent of other steps, **61.4%** of all car trips and **47.5%** of VMT can satisfy this constraint.

Constraint: Walk access distance

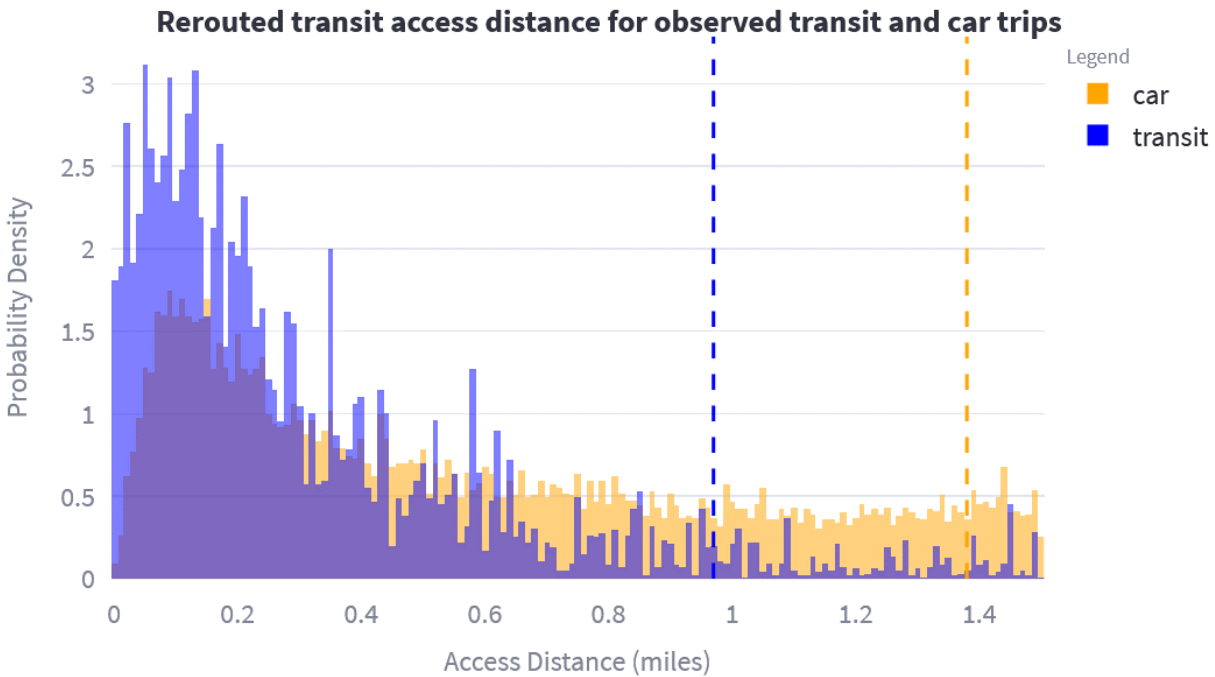
Transit trips are a bit more complicated, because they come in several legs. For example:

- Leg 1: Walk to Bus Stop
- Leg 2: Ride Bus to Another Stop
- Leg 3: Walk to Destination

Here we consider the maximum walking distance to or from a bus stop. Here we consider specifically the walk access distance to a bus stop.

Below is a comparison between the distributions of the distance to a transit stop for observed transit versus that for observed car trips (re-routing data was used for both distributions to allow for a fairer comparison). It can be seen that generally, observed transit trips tend to have lower distances to transit stops, which makes sense as observed transit trips likely find transit attractive in some way, and this is one way it could be attractive to them.

The lines indicate the specified percentile of each of the distributions, and all of the orange distribution to the left of the blue line represent car trips that can feasibly switch to transit, when considering the access distance indicator. **The 95th percentile of observed transit walk access distances falls at 0.97 miles, so we consider it infeasible to walk further than this distance.**



Before this step, **61.4%** of trips could shift to transit feasibly, and after this step, **48.1%** of trips could shift to transit feasibly. Additionally, before this step, **47.5%** of VMT could be mitigated with shifts to transit, and after this step, **37.0%** of VMT could be mitigated with shifts to transit. Overall, independent of other steps, **48.1%** of all car trips and **37.0%** of VMT can satisfy this constraint.

Constraint: Walk egress distance

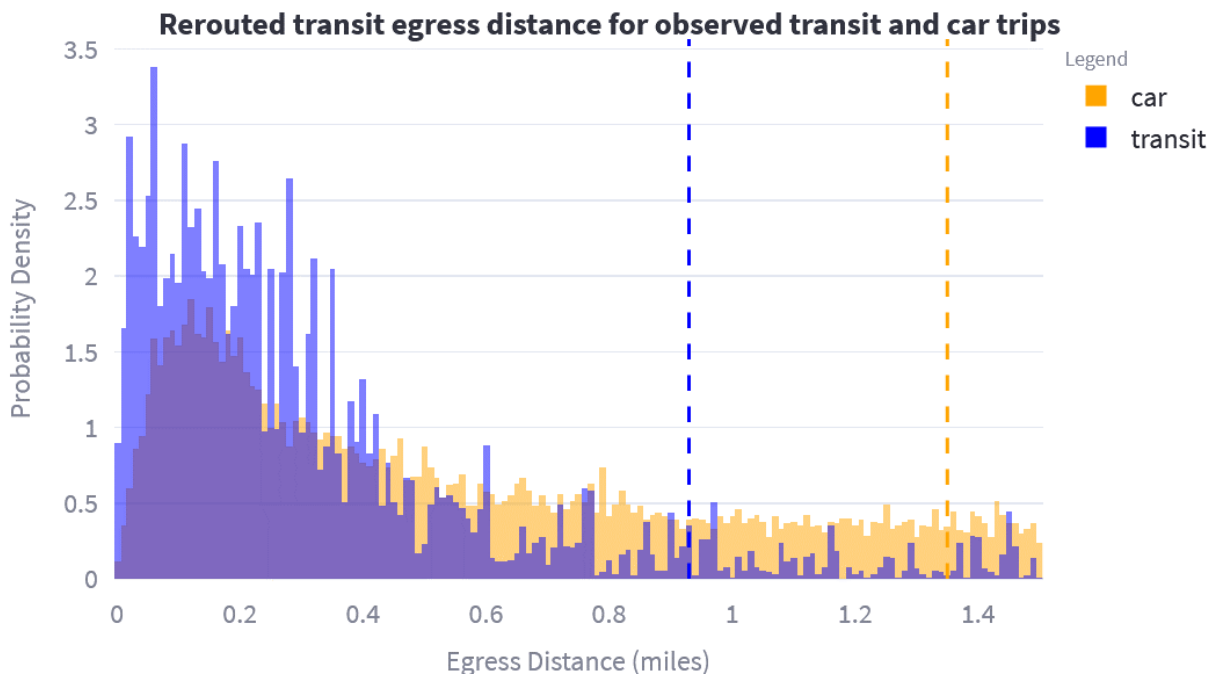
Transit trips are a bit more complicated, because they come in several legs. For example:

- Leg 1: Walk to Bus Stop
- Leg 2: Ride Bus to Another Stop
- Leg 3: Walk to Destination

Here we consider the maximum walking distance to or from a bus stop. Here we consider specifically the walk egress distance to a bus stop.

Below is a comparison between the distributions of the distance from a transit stop for observed transit versus that for observed car trips (re-routing data was used for both distributions to allow for a fairer comparison). It can be seen that generally, observed transit trips tend to have lower distances from transit stops, which makes sense as observed transit trips likely find transit attractive in some way, and this is one way it could be attractive to them.

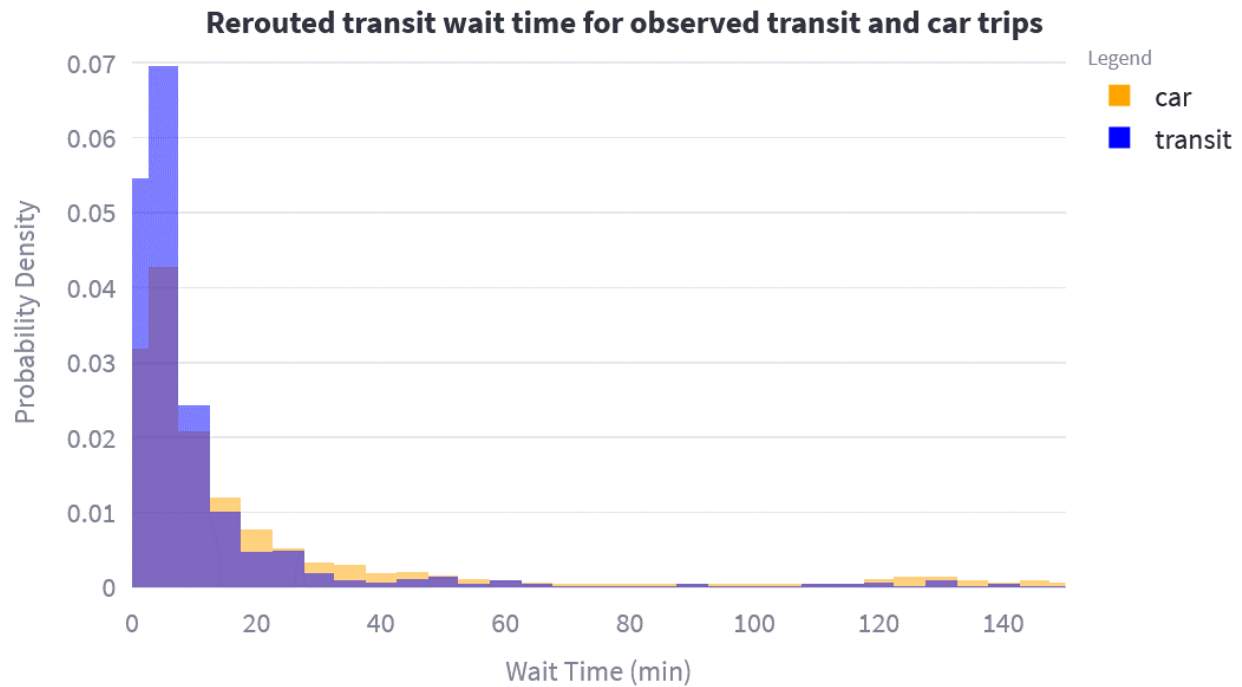
The lines indicate the specified percentile of each of the distributions, and all of the orange distribution to the left of the blue line represent car trips that can feasibly switch to transit, when considering the access distance indicator. **The 95th percentile of observed transit walk egress distances falls at 0.93 miles, so we consider it infeasible to walk further than this distance.**



Before this step, **48.1%** of trips could shift to transit feasibly, and after this step, **39.2%** of trips could shift to transit feasibly. Additionally, before this step, **37.0%** of VMT could be mitigated with shifts to transit, and after this step, **29.1%** of VMT could be mitigated with shifts to transit. Overall, independent of other steps, **48.4%** of all car trips and **36.7%** of VMT can satisfy this constraint.

Constraint: Waiting time

Transit wait time is calculated as the difference between the desired departure time and the departure time needed to catch the transit vehicle. For trips to work and school, we hold the arrival time constant, and the wait time is calculated as the early arrival time. **The 95th percentile of observed transit wait times is 36 minutes, so we consider initial waits in excess of 64 minutes to be infeasible.** The table below shows the distribution of routed waiting times for transit and car trips.

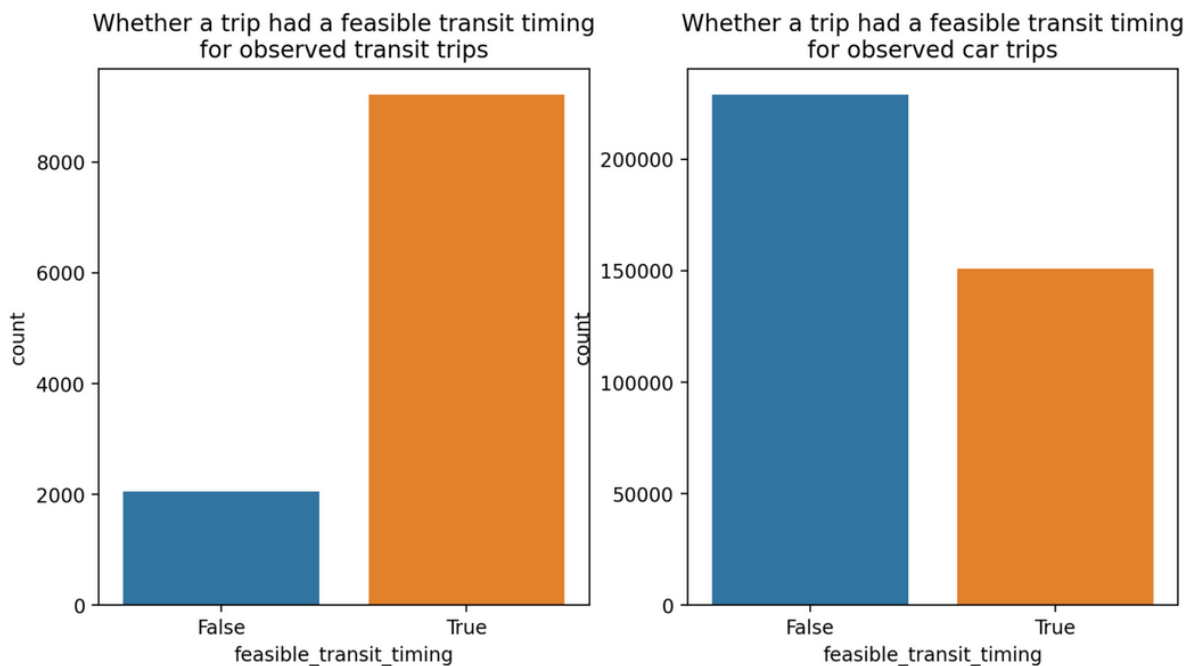


Before this step, **39.2%** of trips could shift to transit feasibly, and after this step, **31.7%** of trips could shift to transit feasibly. Additionally, before this step, **29.1%** of VMT could be mitigated with shifts to transit, and after this step, **21.0%** of VMT could be mitigated with shifts to transit. Overall, independent of other steps, **44.4%** of all car trips and **30.5%** of VMT can satisfy this constraint.

Constraint: Feasible transit timing

For some people, it is infeasible to switch from a car mode to an alternative mode due to timing constraints; with the longer durations of alternative modes, a person may not have the ability to do everything they need to in a given day. **Here we consider whether a trip can be completed by transit within the time available between visiting subsequent out-of-home activities.** With this definition, the first and last trip of the day always have feasible timing because a person can simply depart sooner or arrive later. To provide some flexibility, we allow the traveler to depart up to 5 minutes early and arrive up to 5 minutes late for an out-of-home activity. If a trip cannot be completed by transit within the available time, a traveler would have to reschedule other activities—something which is theoretically possible but beyond the scope of this study.

Here, a bar plot of the result of the timing logic when applied to transit shifts can be seen.



Before this step, **31.7%** of trips could shift to transit feasibly, and after this step, **19.7%** of trips could shift to transit feasibly. Additionally, before this step, **21.0%** of VMT could be mitigated with shifts to transit, and after this step, **12.4%** of VMT could be mitigated with shifts to transit. Overall, independent of other steps, **34.6%** of all car trips and **25.8%** of VMT can satisfy this constraint.